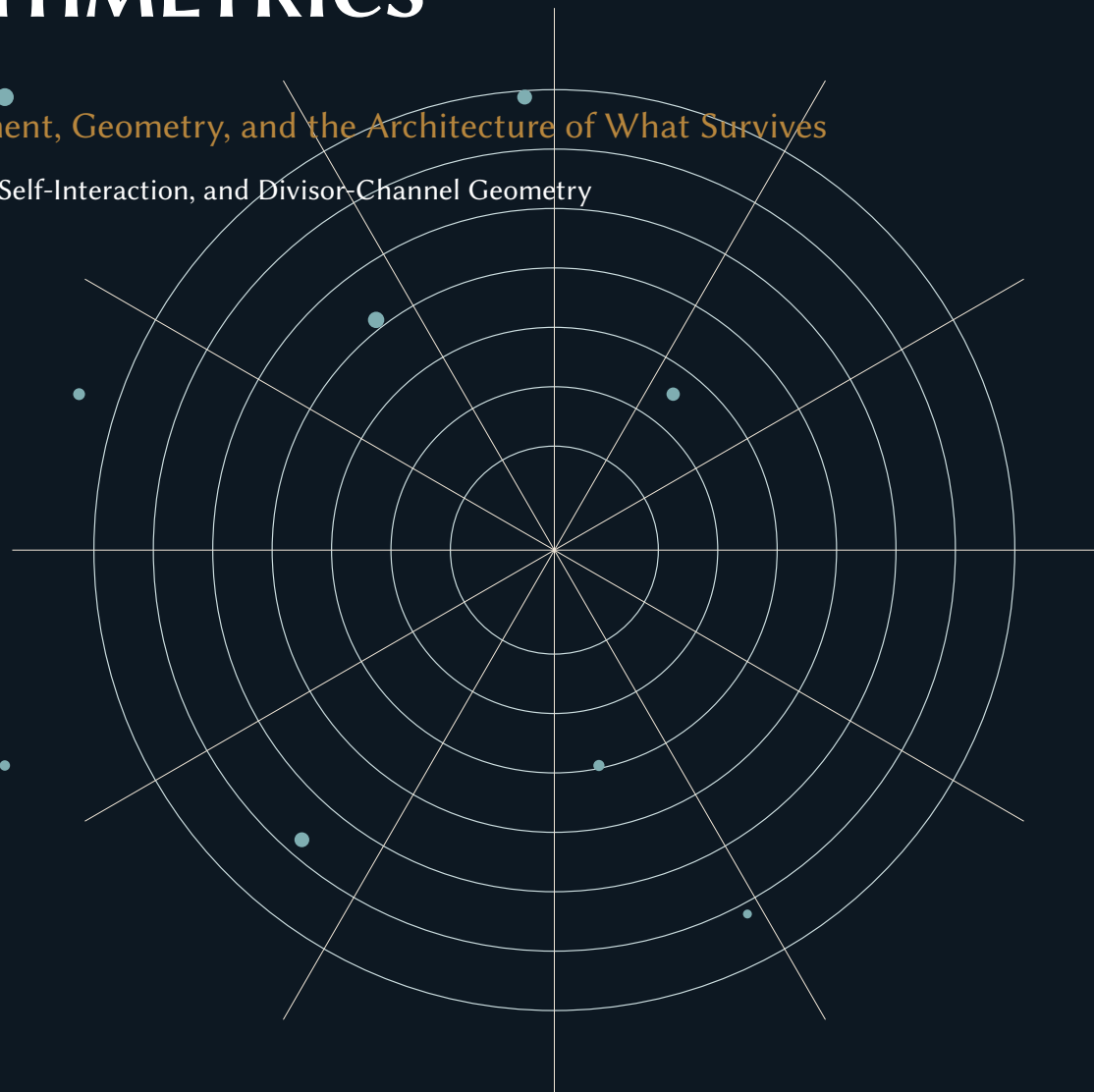


EIGENALGEBRA

ARITHMETRICS

Measurement, Geometry, and the Architecture of What Survives

Perspective, Self-Interaction, and Divisor-Channel Geometry



Thomas Helm

2026

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Mathematical and empirical claims are scoped by the evidence and branch declarations stated in the text.

Preface

This volume brings the revised measurement geometry and the later perspective/self-interaction programme into a single manuscript. The earlier edition established the governing discipline: arithmetic objects become legible only under declared measurement frames; residuals, invariants, and evidence status are part of the object rather than editorial decoration. The enrichment then exposed a consequence that had been present implicitly from the beginning: different structural branches may be exact within their own frames without being universal, and the transport between frames is itself a mathematical object to be typed, measured, and bounded.

The result is not an appendix bolted onto an older book. The material has been renormalized into four continuous movements. Part I develops measurement and perspective. Part II develops the divisor and fibre geometry, including exact closure channels, shadows, and operator resolution. Part III develops self-interaction and structural dynamics before returning to the Jacobi, Toda, fibre-tower, and spectral-circuit machinery. Part IV then states the general theory explicitly, permits bounded non-universal branches, records failure classes, and closes with the falsifiable experimental programme and the question that now governs the work: what survives a change of perspective?

The restraint of the revised edition is retained. A structural resemblance is not an ontological identity. A synthetic PASS is not a theorem, publication authority, or physical claim. A branch earns use by declaring where it lives, what it measures, how it moves, what is invariant, what is invisible to the observer, and what would falsify or bound it.

The architecture of the book is deliberately cumulative. Definitions introduced early are not decorative terminology: they become contracts that later chapters must satisfy. In particular, the distinction between an object, its measured state, its projection, and the residual created by readout is carried forward into the arithmetic, spectral, computational, and governance layers. Readers should therefore treat scope declarations, failure conditions, and evidence labels as part of the mathematics rather than as editorial side notes.

A second recurring discipline is separation of levels. Exact algebraic identities, numerical verification, empirical regularities, architectural conjectures, and physical interpretations are not interchangeable forms of evidence. The text keeps these levels adjacent because their interaction is productive, but it does not allow one level to promote another without an explicit argument. This separation is especially important where familiar language from information theory, spectral theory, topology, or quantum mechanics is used to describe a deterministic arithmetic construction.

A Note on Reading

The book can be read linearly, but it no longer requires a single path. Readers interested in the epistemic architecture should read Chapters 1-7 and then Chapters 26-30. Readers interested in the arithmetic core should continue through Chapters 8-16. Readers interested in self-interaction, cyclic closure, nonlinear channel dynamics, and tiled compatibility should read Chapters 18-20 before the operator machinery of Chapters 21-25.

A productive reading strategy is to distinguish three layers on every page. First identify the declared frame: the domain, observables, and operator class under which the statement is meaningful. Next identify the evidence level: theorem, verification, empirical law, conjecture, or falsification. Finally identify the transport question: what, if anything, survives when the frame changes. This three-pass discipline prevents local exactness from being mistaken for universality and makes the later general theory considerably easier to read.

The notation is intentionally reused only when the underlying typing is compatible. When the same informal word appears in several branches—for example, channel, phase, shadow, rail, or resonance—the surrounding frame declaration determines its precise meaning. Where a transport between meanings is justified, the book states the transducer or commutation relation explicitly; where it is not justified, the resemblance remains an analogy rather than an identification.

The central discipline is frame declaration. A term such as signal, noise, shadow, phase, overlap, channel, resonance, or invariant has no licence outside the measurement plane and operator class that define it. General theory therefore does not mean that every branch applies everywhere. It means that branches share a typed grammar for state, interaction, structural dynamics, projection, residual, failure, and evidence.

Evidence status remains two-dimensional. The first axis asks how well a claim is established: proven, verified, supported, conjectured, or falsified. The second asks where it applies: which branch, frame, parameter regime, boundary condition, and observer. A result may be theorem-grade inside one branch and irrelevant in another.

Legacy theorem labels and named results are retained so that references remain stable across the volume.

Contribution and Evidence Statement

The claims in this volume are classified on two independent axes. The first records evidential status—known result, proved framework result, empirical law, or open programme. The second records scope—the branch, frame, parameter regime, boundary condition, and observer under which the claim is valid. A result may be exact inside one branch and inapplicable in another.

Class K — known results re-derived in the measurement language. These are classical facts used as consistency anchors rather than novelty claims. Representative examples are

$$H_{\text{fact}}(p) = \ln d(p) = \ln 2 \quad (p \text{ prime}), \quad (0.1)$$

$$\Delta S_{\text{mul}}(n, m) = 0 \iff \gcd(n, m) = 1, \quad (0.2)$$

$$\sum_{j=0}^{N-1} \ln \frac{j+2}{j+1} = \ln(N+1), \quad (0.3)$$

$$d(n) \text{ odd} \iff n \text{ is a perfect square}, \quad (0.4)$$

$$\lim_{N \rightarrow \infty} \frac{1}{N^2} \#\{(n, m) \in [1, N]^2 : \gcd(n, m) = 1\} = \frac{6}{\pi^2} = \prod_p (1 - p^{-2}), \quad (0.5)$$

$$\sum_{n \geq 1} \frac{d(n)}{n^s} = \zeta(s)^2 \quad (\Re s > 1). \quad (0.6)$$

Class N — new results with proof inside the framework's definitions. The volume records, among others, the squarefree CV product formula

$$1 + \text{CV}(N)^2 = \prod_{p|N} \left[1 + \left(\frac{p-1}{p+1} \right)^2 \right] \quad (N \text{ squarefree}), \quad (0.7)$$

the fixed-divisor/index parity correspondence, and the refined skip-set criterion developed in Chapter 14. The arithmetic uncertainty construction is stated there with its observables made explicit; its lower bound is a claim about those observables, not a frame-independent uncertainty principle.

Class E — empirical laws with statistics. The principal empirical claims include the fitted relation

$$\beta_{\text{eff}}(N) \approx 1 - e^{-a \text{CV}(N)}, \quad a \approx 0.88, \quad (0.8)$$

with reported Pearson correlation $r = 0.905$ and RMSE 0.049 on the stated CIM ensemble; the entropy-domain transition between fibres with divisor counts 6 and 7; the phase-disorder fit $\beta_{\text{eff}} \sim \varepsilon^{0.21}$; and the reported round-trip-residual and theta-oracle benchmarks. These are programme measurements, not theorem-level statements. Where the adjacent-gap ratio is used, it is written $\langle r \rangle$ and is not identified with β_{eff} .

Class O — open conjectures and programmes. These include the automorphic-ladder top rung, the annular spectral waveguide, the theta-function gap-collapse oracle, higher prime-rail iterates, and computational-prime certification. Their promotion bars are collected in Appendix D.

The reader should hold this classification against every later chapter. A synthetic PASS validates a declared criterion for a declared model; it is not by itself a theorem, publication authority, production authority, or physical claim.

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Part I

Measurement, Perspective, and the Observed Object

Measurement is not an afterthought. Perspective is part of the state that becomes legible.

1. Arithmetic and Arithmetics

Arithmetic tells us what numbers do. The discipline developed in this book asks what numbers are when measured. The distance between those questions is one suffix wide and one orientation deep. The distinction can be felt in a simple example. Take the integer twelve. Arithmetic knows that twelve is three times four, that it is divisible by two, three, four and six, that its prime factorisation is $2^2 \cdot 3$. These are facts about operations and their results. They are correct and useful. They do not yet tell us how twelve sits relative to other integers when both are regarded as structured objects carrying internal composition. Once we choose to measure that internal composition, new quantities appear — and they appear not as decorations on twelve but as properties of twelve-under-a-protocol. The number of positive divisors of twelve is six. The natural logarithm of six is a measure of the information carried by the fibre of solutions that map to twelve under a simple projection. When two integers are multiplied, the change in that informational quantity vanishes precisely when the integers are coprime: $\ln d(3 \cdot 4) = \ln 6 = \ln 2 + \ln 3 = \ln d(3) + \ln d(4)$, because $\gcd(3, 4) = 1$; but $\ln d(2 \cdot 6) = \ln 6 \neq \ln 2 + \ln 4 = \ln d(2) + \ln d(6)$, because the shared factor 2 collapses divisor combinations that the naive product count expects. Two distinct primes always stand at the same angular separation in a geometry whose distance is derived from the overlap of their divisor structures. These are not analogies. They are exact statements about measured arithmetic objects under a declared protocol. The protocol is essential, and it is worth saying precisely what one is. A measurement frame declares: the domain (what raw objects are admitted), the units (what counts as magnitude), the observables (which functions of the raw object are computed), the coordinate chart (how those computations are assembled into a state), and the null model (what the frame would report on structureless input). Without all five, the question what is twelve? is underdetermined. Under one frame twelve carries fibre entropy $\ln 6$ and a specific spectral signature — the eigenvalues of a Jacobi matrix built from its six divisors. Under another frame — say, the prime rail of Chapter 11 — twelve is filtered out entirely: it does not exist as a rail element. The integer has not changed. The conditions of legibility have changed. The object that arithmetic takes for granted becomes, under measurement, a state relative to a frame. This is the first and most important commitment of the book. Measurement is not a secondary act performed on a pre-existing object. Measurement is the act that renders an object legible as a state. Different frames can produce different states from the same raw integer without contradiction, because the state includes the frame. Operational realism of this kind is ordinary in experimental science — no physicist confuses a thermometer reading with the temperature-in-itself — and strangely rare in the elementary treatment of number, where the value on the line is routinely taken to exhaust what the number is. The geometry that follows is the systematic development of that ordinary insight inside arithmetic. The name given to the resulting discipline is arithmetics. The word is intentionally close to arithmetic. The difference is a single suffix and an entire orientation. Arithmetic operates. Arithmetics measures. The number line remains available as one coordinate chart among others. It is no longer the only chart, and it is no longer the definition of what a number is. From this distinction everything else in the book unfolds: the observer treated as an operator rather than a subject (Chapter 2), the zero mode understood as identity-like degeneracy rather than vacuum (Chapter 4), the chiral structure that records orientation under measurement (Chapter 5), the residual that quantifies what readout destroys (Chapter 16), and finally the canonical measured object that carries all of these features at once (Chapter 7). The reader is invited to hold the distinction lightly at first. It will be tested, sharpened, and given formal expression in the chapters that follow. If the distinction fails, the rest of the architecture fails with it. If it holds, a coherent geometry of measured arithmetic objects becomes possible.

2. Measurement Before Ontology

The conventional picture of observation places the observer outside the system. A subject looks in, records what is already there, and withdraws without consequence.

The object is presumed to exist in a definite state independent of the looking. Measurement, on this view, is an epistemic inconvenience that imperfectly reveals a prior ontology. The geometry developed here inverts the order. The object is not given. The object is what survives the measurement that makes it legible. Remove the measurement protocol and the object does not persist invisibly as the same thing; it persists as something else, measured differently, or not at all. Being, for the purposes of this book, is coordinatestable recurrence under admissible perturbation of the measurement frame. This is not a claim about the ultimate furniture of the world. It is a claim about the only objects the present geometry is competent to discuss. The geometry does not deny that unmeasured reality may exist. It simply refuses to speak about it. What it can speak about are states that have been rendered legible by a declared protocol and that remain coherent when the protocol is varied within controlled limits. The refusal is not modesty performed for effect; it is a competence boundary, and like every competence boundary in this book it is load-bearing. A discipline that cannot say what it cannot say will eventually say anything. The technical correlate is straightforward. A constraint cell consists of a solution space X , a set of constraints E , a projection $\pi : X \rightarrow \mathbb{Z}$ onto the integers (or another declared domain), and a domain declaration D . The fibre of solutions assigned to a particular integer, $F_n = \pi^{-1}(n)$, is not an intrinsic property of that integer considered in isolation.

It is a measured property relative to the cell. Change the projection or the constraints and the fibre changes. The integer remains; its measured structure does not. The observer is therefore not a subject standing outside. The observer is the measurement protocol itself — the aperture through which legibility occurs. Certification is stronger than observation, and the gap between them is where most of the book's machinery lives. To observe is to project through a coordinate system. To certify is to show that the resulting state survives across coordinate systems, that its identity persists when the measurement frame is perturbed. A candidate may appear stable in one coordinate while failing in another. Such a candidate is observed but not certified. Certification requires cross-coordinate coherence, and the geometry therefore refuses single-coordinate ontologies.

The running example makes this concrete. Seven is prime is true and useful under the ordinary value frame. Under the prime-rail frame — where the observable is ordinal position in the sequence of primes — seven is p_4 , and the index $4 = 22$ is composite. The object has not changed. The conditions of legibility have changed. A single-coordinate account must choose which seven to believe; a multi-coordinate account records both facts without contradiction, and the enriched state of Chapter 11 carries value-sector and index-sector simultaneously. The enrichment is not redundancy. It is the price of refusing to mistake one chart for the territory. Two overclaims are rejected at the outset, because experience shows they are the two most seductive. The first is the claim that structural resemblance to quantum formalism implies that the system is quantum. It does not. The operators employed are classical numerical linear algebra. The grading is combinatorial. The bounds are arithmetic facts about divisor counts. No superposition of physical states is present, no Born rule governs outcomes, no experimental prediction about physical systems is made. The safer and more accurate framing is deterministic measurement geometry. The second overclaim is the elevation of a geometric singularity — the zero mode — into a metaphysical ground of being. The zero mode is the point where a coadjoint orbit collapses and orientation becomes undefined. It marks a limit of the measurement frame, not the foundation of existence. Both refusals are architectural. They prevent the inflation of a precise formal construction into a totalising narrative, which is the failure mode that destroys frameworks of this kind from the

inside. The measurement turn is therefore an epistemological inversion with technical consequences. Ontology asks what is. The present discipline asks what survives measurement. The second question is answerable with the apparatus at hand. The first is not, and the book does not pretend otherwise. We begin with protocols and recognise as objects only what those protocols render stable. The rest is noise, artefact, or candidates awaiting better frames.

3. Perspective, Projection, Shadow, and Common Mode

Frame declaration. This chapter treats perspective as an operator, not as a subjective metaphor. A “plane” is a declared measurement map or projector. Signal, noise, shadow, overlap, and interference are therefore properties of a state relative to that map and objective.

Let H denote latent structure and let Π_P be the measurement associated with perspective P . The observed state is

$$y_P = \Pi_P(H).$$

The same H may produce incompatible-looking but simultaneously correct observations under P_1 and P_2 because the measured state includes the frame.

Definition 3.1 (Shadow). *The shadow of H on plane P is*

$$\text{Shadow}_P(H) := \Pi_P(H).$$

A shadow is therefore not the latent object. It is the latent object after a declared projection geometry.

3.1. Six canonical measurement planes

The framework uses six recurring planes.

Raw incidence plane: preserves declared support membership.

Normalized-overlap plane: rescales overlap by source, target, or joint capacity.

Common-mode-subtracted plane: removes structure shared by all admissible states.

Phase-sensitive plane: retains signed or complex amplitudes, permitting constructive and destructive interference.

Spectral plane: resolves states in an eigenbasis or character basis.

Quotient plane: identifies states equivalent under the declared observer and records only the quotient class.

The universal divisor 1 is the canonical example. On the raw divisor-incidence plane it is genuine common structure. On a prime-relative normalized plane it contributes a permanent one-half overlap floor. On a differential plane that quotients or subtracts the universal component, it disappears. None of these statements contradicts the others.

3.2. Signal and noise are task-relative

Suppose $H = S \oplus B \oplus R$, where S is target structure, B is a persistent background, and R is residual variation. One observer may measure B and therefore call it signal. Another may seek S and therefore treat B as nuisance. A third may use B as a calibration reference. The correct formal question is not “is B noise?” but

what information does $\Pi_P(B)$ carry about the declared target?

A deterministic common mode can be structurally real and discriminatively useless at the same time.

3.3. Interference requires amplitude structure

Set overlap alone cannot be destructively interferometric because cardinalities are nonnegative. Destructive interference becomes meaningful only after a signed or complex amplitude is declared. Let each channel d carry

$$z_d = w_d e^{i\phi_d}.$$

Then a phase-sensitive observation has the form

$$Z_P(n) = \sum_{d|n} w_P(d) e^{i\phi_P(d)}.$$

Channels reinforce when phases align and cancel when phases oppose. Thus “shadow”, “background”, and “destructive interference” are different readings of the same structural content under different algebras of observation.

3.4. Ontic and perspectival invariants

Within this framework an **ontic invariant** is reserved for a quantity proven invariant across every admissible plane in a declared family. A **perspectival invariant** is stable only inside a specified plane or compatible class of planes. Most quantities in Arithmetics are perspectival by construction. The framework therefore treats frame-independence as something to prove, not something implied by elegance.

Bounds and portability. No claim is made that all possible perspectives can be represented by linear projectors. The operator notation is the canonical branch for the present deterministic measurement geometry. Nonlinear observers are admitted as separate branches with their own transport laws.

Failure condition. Calling a component “noise” without stating the target and observer is a semantic failure. Treating a null projection as proof of latent absence is an observability failure. Calling nonnegative overlap counts “destructive interference” without a signed or phase-sensitive amplitude is an algebra mismatch.

4. The Zero Mode

Every coadjoint orbit of a compact group carries a geometry. In the simplest interesting case — the orbits of $SU(2)$ through its dual Lie algebra — the orbits are spheres whose radius is fixed by a Casimir invariant. As long as the radius is positive the orbit is a smooth manifold, a symplectic form equips it, and a natural dynamics can traverse it. At radius zero the sphere collapses to a single point. That point is the zero mode. The zero mode is not a pathology of the formalism. In the Lie–Poisson picture it is the origin of the dual Lie algebra, where the coadjoint orbit has collapsed to a singleton. It is not the identity element of the group itself. At that point the symplectic form is trivial because the orbit has zero dimension, the orientation that elsewhere distinguishes chiral sectors becomes undefined, and there is no nontrivial orbit left to traverse. Dynamics that depend on a nonvanishing orbit simply stop. The stoppage is correct behaviour, not numerical failure. In the arithmetic setting the zero mode first appeared as an inconvenience. Certain iterative spectral procedures lost resolution when matrices approached the zero configuration. Eigenvalues compressed. Level statistics drifted toward degeneracy. The machinery appeared to fail, and the first response was to hunt for the bug. The resolution was geometric, not computational. The fixed point of the dynamics occurs precisely when the matrix sits at the centre of the coadjoint orbit system — that is, when the orbit has degenerated to a point. There is nothing left to evolve. The apparent saturation was the correct signature of arrival at the degenerate orbit. This episode is recorded in the ledger as a methodological lesson: saturation of an iterative spectral procedure is a signal to be interpreted before it is a defect to be repaired. Any distinguishable state must occupy a non-zero orbit and therefore carries a positive Casimir value. The smallest such value that the present geometry can produce for an arithmetic object is fixed by the fibre entropy of a prime. Every prime has exactly two positive divisors and therefore the same informational content under the natural measurement frame: $H_{\text{fact}}(p) = \ln 2$. The natural logarithm of two is the minimal informational stretch away from the zero mode that produces a recognisable object. It functions as a Landauer-type floor: the least information required to separate one state from the degenerate point. All primes share this floor because all primes share the same elementary divisor structure. The constancy of the floor is also the geometric reason that any two distinct primes stand at a fixed angular separation in the Fisher geometry of divisor states — the subject of Chapter 14.

Ledger note — on the name. The term “Landauer floor” is retained from the development history, but it is an analogy of form, not a claim of physics. Landauer’s principle bounds the energy cost of erasing a bit in a physical system at temperature T . Nothing here is a physical system, nothing is at any temperature, and $\ln 2$ enters through divisor cardinality, not through thermodynamics. The name marks the structural role — a minimal distinguishability cost from which composite costs are built — and no more. Where the name threatens to import physics, read “minimal fibre entropy” instead.

The zero mode is therefore the reference against which measurable arithmetic objects are distinguished. It is an identity-like degeneracy inside a deterministic measurement geometry. It is not a physical vacuum. It is not a zero-point energy. It is not a metaphysical ground of being. These refusals are repeated because the temptation to inflate the singularity is persistent, and because the inflation, once admitted, propagates: a zero mode promoted to a ground of being silently rewrites every residual bound that references it, converting measurement statements into cosmological ones. The book treats the temptation as a failure mode and declines it. From the zero mode one can stretch into distinguishable states. The geometry of that stretching, the informational cost of the first step, and the orientation that becomes available once the orbit is non-degenerate, occupy the next chapters. The zero mode itself

remains the silent origin: the point at which measurement has not yet begun because there is nothing left to measure.

5. Orientation and the Chiral Structure

Once a non-degenerate orbit is present, orientation becomes available. The arithmetic branch introduces a $\mathbb{Z}_2$ grading and a Dirac-type operator that exchanges the two graded sectors. The cleanest invariant in this branch comes from the divisor involution

$$\delta \mapsto \frac{n}{\delta}.$$

A fixed point satisfies $\delta = n/\delta$, hence $\delta^2 = n$. Therefore a fixed point exists exactly when n is a perfect square, and in that case the unique fixed divisor is $\sqrt{n}$. Equivalently, $d(n)$ is odd exactly for squares because all non-fixed divisors occur in complementary pairs.

For the alternating grading used by the divisor-indexed Jacobi chain, the index is 1 when $d(n)$ is odd and 0 when $d(n)$ is even. Thus, within this branch, three indicators coincide:

$$\# \text{Fix}(\delta \mapsto n/\delta) = \text{ind } D_n = \mathbf{1}_{\text{sq}}(n).$$

This is the self-dual divisor/index correspondence developed in Chapter 14. Calling the protected zero mode “Majorana-type” is an analogy to a self-conjugate chiral zero mode; it is not a claim about physical Majorana particles.

The grading also supplies an orientation-reversing algebraic conjugation. Its role is structural: it compares the two graded sectors and provides a natural way to formulate reversal tests. It does not, by itself, imply that future states can be inferred from past states. Such a claim requires the separately declared dynamics and readout map.

This distinction matters for irreversibility. Exact Toda transport is isospectral and reversible on its domain of existence. The readout family of Chapter 14 attenuates off-diagonal information for $0 < \varepsilon < 1$ but remains invertible in exact arithmetic; genuine algebraic information loss appears at the singular projection $\varepsilon = 1$ or when another non-invertible measurement operation is explicitly introduced. The reported round-trip residual correlation is therefore an empirical diagnostic of the declared pipeline, not a general proof that every nonzero attenuation is irreversible.

The orientation, index, zero-mode, and residual constructions are thus kept in one typed package without physicalising them. Perfect-square detection is an exact arithmetic statement in the declared divisor/Jacobi branch. Stronger claims about programme-specific spectral-curve genus or physical duality require their own definitions and evidence and are not used to establish the square-detection result.

6. Geometry as the Residue of Constraint

Geometry, in the sense used here, is the record of a restriction. A larger space is constrained; the residue of the constraint is what remains available for measurement. There is no free-floating form that is later bound. There is constraint, and the form is what survives it. Four primary substrates appear repeatedly, and it is worth naming them as a system, because every later construction is built from exactly these four and nothing else. The first substrate is spectral. Eigenvalues and level statistics arise under isospectral dynamics that preserve certain invariants while allowing the frame to rotate. The second substrate is connective. Parallel transport and holonomy record how structure is moved from one place to another without tearing. The third substrate is invariant. Casimirs, indices, residuals, fibre entropies and Fisher distances are quantities that survive the allowed transformations. The fourth substrate is algebraic. The operators that generate the measurable quantities — Jacobi matrices, Dirac-type operators, Cartan projections — form the concrete language in which the geometry is written. Each substrate is the geometric record of a constraint, and the elementary instance is the constraint cell itself. A solution space is restricted by a constraint set and projected onto a declared domain. The fibre that results is the measurable object. Change the constraint or the projection and the fibre changes. The geometry does not float free of the restrictions that produce it. It is those restrictions made visible and measurable. This way of thinking has a developmental history inside the work that produced the present book, and the history is recorded because the order of discovery is itself instructive. Fibre cardinality was first recognised as information. The Fisher metric on divisor states was shown to be exact. The zero mode was reinterpreted as identity-like degeneracy rather than numerical pathology. The Cartan projection was isolated as the sole irreversible step in an otherwise reversible dynamics. The perfect-square index was identified as a topological fact of the grading. Only then was the seven-component measured state assembled as the canonical object that carries all of these features at once. The sequence is recorded so that the reader can see that the residual accounting was itself discovered, not stipulated. A reader tempted to skip to Chapter 7 should know that the canonical object looks arbitrary when encountered first and inevitable when encountered last.

Cultural and historical motifs appear in later chapters. They are admitted only as pattern sources — suggestive generators of computational primitives. They are never offered as evidence that earlier cultures possessed the modern formalism. The distinction is strict and is maintained throughout: a pattern source licenses a construction; it does not certify a lineage.

7. The Spectral Observer-Eigenflow State

The preceding chapters have prepared the ground. It is now possible to define the canonical measured object of the entire discipline.

A raw object — an integer, a candidate feature, a computational species, a constraint cell — carries no intrinsic coordinates. It is the pre-measurement substrate. Once a measurement frame is declared, the raw object becomes a measured state. The frame specifies domain, units, observables, coordinate chart, null model and governance rules. The same raw object under different frames yields different states. The frame is part of what is measured. The measured state is given coordinates. Five have proved necessary and, so far, sufficient — the Eigenform Coordinate Pentad: Angular coordinate — records geometric position, overlap and Fisher distance. Functorial coordinate — records how the state behaves under admissible transformations of frame or substrate. Informational coordinate — records fibre entropy and its changes. Monoidal coordinate — records compositional type: whether the state factorises cleanly or carries residual interaction. Temporal coordinate — records eigenflow: trajectory, recurrence, holonomy, stabilisation or drift under the allowed dynamics. A genuine eigenform must exhibit coherence across the pentad. Stability in a single coordinate is observation; multi-coordinate survival under residual-checked perturbation is certification. This is the formal content of the observation–certification gap introduced in Chapter 2, and the defect machinery below exists to make failures of pentad coherence explicit rather than impressionistic. Higher structure is then attached. The derivative jet records how each coordinate changes when the frame, the substrate or the governance rules are perturbed. The defect vector records typed instability across the five coordinates. A candidate may be temporally recurrent yet functorially incoherent; the defect vector makes the failure explicit. The gate set records the governance conditions under which the candidate is released, quarantined, restricted or rejected. Residual magnitude and defect magnitude enter the gate decisions. Finally an evidence ledger records the history of claims made about the state. The ledger is append-only. The complete package is the Spectral Observer-Eigenflow State:

$$\text{SOES}(a) = (a, F, A(a), Jk A(a), D(a), G(a), L(a))$$

raw object, frame, coordinate pentad, derivative jet, defect vector, gate set, ledger. It is the object that the rest of the book elaborates. Every subsequent construction is required to be expressible as a statement about such a state or about a residual of such a state. No formal object is admitted that cannot meet this reduction requirement. The requirement is the book's main defence against vocabulary drift: a term that cannot be reduced to pentad coordinates, residuals, gates or ledger entries is not a term of the geometry, however evocative it may be. The definition is deliberately austere. It does not claim that every possible measurement geometry must take this form. It claims only that the geometry developed in this book does take this form, and that the form is coherent enough to support the operator theory, the contact geometry, the computational realisations and the governance discipline that follow. Whether richer or leaner forms are possible is an open question. The present form is the one that has survived the residual accounting of the programme that produced it. With the canonical object in hand, the book turns outward. Part II develops the arithmetic constructions that motivate the whole geometry: constraint cells and fibre entropy, divisor-lattice geometry, observer bounds, the measurement axis, and the computational morphospace. Part III turns to operators, towers and computation. Throughout, the tone remains deliberate: technical detail where precision requires it, philosophical commentary where the technical detail would otherwise be misread as metaphysics. The reader who has followed the argument this far already possesses the generative core. Everything else is expansion, specialisation, or caution.

Part II

Constraint, Fibre, and Arithmetic Geometry

The arithmetic branch is exact where its frame is exact, and bounded where the frame ends.

8. The Layer Architecture

The Cellular Integer Framework (CIF) is organised as a dependency stack. Each layer adds observables or operators while retaining the scope declarations of the layers below it.

Layer I – Constraint cells. A base cell is

$$C = (X, E, \pi, D), \quad F_n = \pi^{-1}(n), \quad (8.1)$$

where X is a solution space, E a constraint set, $\pi : X \rightarrow \mathbb{Z}$ a projection, and D a domain declaration.

Layer II – Fibre entropy. Six families assign log-cardinality or signed log-cardinality observables to declared fibres. They are not Shannon entropies unless a probability measure is separately specified.

Layer III – Hybrid arithmetic and geodesic chains. The branch uses the declared curvature observable

$$\kappa_H(a, b) = 2^{-1/2}(a + b) \quad (8.2)$$

and the skip set

$$S = \{(n, m) : \gcd(n, m) = 1\}, \quad (8.3)$$

whose asymptotic density is $6/\pi^2$.

Layer IV – k -partite Coloured Integer Matrix algebra. A CIM is a $V_n \times V_n$ integer matrix on the balanced complete k -partite graph $K_{n, \dots, n}$, with $V_n = kn$, zero diagonal blocks, and arbitrary off-diagonal blocks. The graph automorphism group is

$$G_c \cong S_n \wr S_k, \quad |G_c| = (n!)^k k!. \quad (8.4)$$

The Toda Lax flow $\dot{L} = [B, L]$ preserves spectral invariants $I_j = \text{tr}(L^j)$ when the stated Lax-pair assumptions hold.

Layer V – Zeta families. Ten Dirichlet-type series encode distinct projections of the lower layers. The fibre-cardinality family

$$F(r, s) = \sum_{n \geq 1} d(n) r n^{-s} \quad (8.5)$$

satisfies $F(0, s) = \zeta(s)$ and $F(1, s) = \zeta(s)^2$ for $\Re s > 1$.

Layer VI – Random-matrix statistics. The factor-fibre ensemble is analysed using both a fitted repulsion coordinate β_{eff} and, where explicitly stated, the adjacent-gap ratio $\langle r \rangle$. These are separate observables. Standard reference values are approximately $\langle r \rangle = 0.386$ (Poisson), 0.536 (GOE), and 0.603 (GUE).

Layer VII – Information geometry. Divisor states induce the angular metric defined in Chapter 14. Distinct primes are equidistant under that metric because their normalized divisor states share only the unit divisor.

Layer VIII – Divisor-state lifting. Each integer is mapped to the unit vector

$$|\psi_N\rangle = \frac{1}{\sqrt{d(N)}} \sum_{k|N} |k\rangle. \quad (8.6)$$

The terminology is algebraic: no physical quantum state is asserted.

Layer IX – Automorphic programme. A sequence of fibre Dirichlet series links the baseline zeta function to entropy-weighted, character-twisted, and Gaussian constructions. The proposed top rung remains open.

8.1. The ten zeta families

Layer V defines ten zeta families, each capturing a different projection of the framework's arithmetic geometry. They are listed here with their identifying fibres. The automorphic programme uses the r -derivative of the fibre-cardinality family as a derivation.

Table 8.1.: *The ten zeta families of Layer V.*

Series	Fibre / role
$\zeta_{\text{CIF}}(s) = \sum_j \lambda_j^{-s}$	CIM spectral
$\zeta_{H,\text{spec}}(s) = \sum_j H_{\text{fact}}(\lambda_j) \lambda_j^{-s}$	Entropy-weighted spectral
$\zeta_H(s) = \sum_{n \geq 1} H_{\text{fact}}(n) n^{-s}$	Factorisation entropy
$\zeta_B(s) = \sum_j B_H(j) j^{-s}$	Holonomy budget
$F(r, s) = \sum_{n \geq 1} d(n)^r n^{-s}$	Fibre cardinality
$\zeta_G(s) = \sum_{n \geq 1} H_G(n) n^{-s}$	Gaussian fibre
$\zeta_{\text{sf}}(s) = \sum_{n \text{ squarefree}} H_{\text{fact}}(n) n^{-s}$	Squarefree fibre
$\zeta_{H,\chi}(s) = \sum_{n \geq 1} \chi(n) H_{\text{fact}}(n) n^{-s}$	Character-twisted fibre
$\zeta_{\text{cl}}(s) = \sum_D h(-D) n^{-s}$	Class-number fibre
$\zeta_{\text{CIF}}(s, t) = \sum_j \lambda_j(t)^{-s}$	Flow-parameter spectral

These families are not a taxonomy for its own sake. The automorphic ladder uses the ordinary derivation $\partial_r|_{r=0}$ on $F(r, s)$ and its twisted analogues. A differential graded algebra would require additional grading and nilpotence data; no such structure is claimed here. The universal ratio theorem (T05) states that for any prime p and any s with $\Re(s) > 0$, the ratio $\zeta_{\text{CIF}}(s)/\zeta_H(s)$

evaluated on $K_{p,p}$ equals $\ln 2$ — a fixed point linking the spectral and entropy layers.

An honest obstruction belongs here. The programme does not presently retain a validated closed-form zero locus for the CIM spectral zeta. Consequently no claim is made that this zeta-family formalism provides an approach to the Riemann hypothesis. The framework's zeta functions are bookkeeping devices for their own fibres; where they coincide with classical L -functions (Chapter 21), the coincidence is stated only at the evidence level actually established.

9. Fibre Entropy Families

The framework uses six fibre-entropy families on $\mathbb{Z}_{\geq 1}$. Each family measures a declared algebraic fibre; changing the fibre changes the observable.

9.1. The factorisation fibre

The foundational observable is

$$H_{\text{fact}}(n) = \ln d(n), \quad (9.1)$$

where $d(n)$ is the number of positive divisors. For a prime p ,

$$H_{\text{fact}}(p) = \ln 2 =: \lambda_{\min}. \quad (9.2)$$

This common value reflects the two-element divisor set $\{1, p\}$. In the tensor encoding used later, that two-state choice is represented by a vector space $C_2 \cong \mathbb{C}^2$; it is *not* the dimension of an irreducible representation of S_2 .

For the running example,

$$12 = 2^2 \cdot 3, \quad d(12) = (2+1)(1+1) = 6, \quad H_{\text{fact}}(12) = \ln 6. \quad (9.3)$$

9.2. The composition fibre

The second family is

$$H_{\text{comp}}(n) = \ln(n+1). \quad (9.4)$$

The count $n+1$ is the number of ordered pairs $(a, b) \in \mathbb{Z}_{\geq 0}^2$ with $a+b=n$. It is not the number of partitions of n into at most two parts. This correction changes the combinatorial interpretation, not the telescoping identity in Theorem T02.

9.3. Gaussian, Eisenstein, squarefree, and character fibres

For the Gaussian branch define

$$d_{\mathbb{Z}[i]}(n) = \#\{\alpha \in \mathbb{Z}[i] / \sim : \alpha \mid n\}, \quad H_G(n) = \ln d_{\mathbb{Z}[i]}(n), \quad (9.5)$$

where $\sim$ identifies associates. For a rational prime p this gives

$$d_{\mathbb{Z}[i]}(p) = \begin{cases} 3, & p = 2 \quad (\text{ramified}), \\ 4, & p \equiv 1 \pmod{4} \quad (\text{split}), \\ 2, & p \equiv 3 \pmod{4} \quad (\text{inert}). \end{cases} \quad (9.6)$$

The ramified value is 3 because, up to associates, 1, $1+i$, and 2 all divide the rational integer 2. This is the standard divisor-class count implied by the definition above.

The resulting Dirichlet series over rational integers has Euler product

$$\sum_{n \geq 1} \frac{d_{\mathbb{Z}[i]}(n)}{n^s} = \zeta(s)^2 (1 + 2^{-s}) \prod_{p \equiv 1 \pmod{4}} \frac{1 + p^{-s}}{1 - p^{-s}}, \quad \Re s > 1. \quad (9.7)$$

At $s = 2$ its value is approximately 3.76676.

For the Eisenstein ideal-counting branch,

$$F_{\mathbb{Z}[\omega]}^{\text{id}}(1, s) = \zeta(s) L(\chi_{-3}, s), \quad (9.8)$$

with the empirical validation range stated later in Chapter 21. The squarefree and character-weighted families are

$$H_{\text{sf}}(n) = |\mu(n)| H_{\text{fact}}(n), \quad (9.9)$$

$$H_{\chi_{-4}}(n) = \chi_{-4}(n) H_{\text{fact}}(n). \quad (9.10)$$

On squarefree $n = \prod_{j=1}^r p_j$, $H_{\text{sf}}(n) = r \ln 2$.

Six families therefore instantiate one principle: fibre entropy is a measurement of a declared solution space, not an intrinsic scalar attached to an integer independently of frame.

10. The Core Theorems

Three core results anchor the arithmetic entropy layer. T02 and the elementary part of T19 are direct identities. T20 packages standard Gaussian-prime splitting together with the framework's chosen divisor-count observable.

10.1. T02: Phase-space telescoping

Define

$$\Delta(j) = \ln \frac{j+2}{j+1}. \quad (10.1)$$

Then

$$\sum_{j=0}^{N-1} \Delta(j) = \ln(N+1) = H_{\text{comp}}(N). \quad (10.2)$$

The proof is immediate by telescoping. The identity is exact and carries no truncation or asymptotic remainder.

10.2. T19: Entropy production formulas

For positive integers n, m define

$$\Delta S_{\text{add}}(n, m) = \ln(n+m+1) - \ln(n+1) - \ln(m+1) \leq 0, \quad (10.3)$$

$$\Delta S_{\text{mul}}(n, m) = H_{\text{fact}}(nm) - H_{\text{fact}}(n) - H_{\text{fact}}(m) \leq 0. \quad (10.4)$$

Moreover,

$$\Delta S_{\text{mul}}(n, m) = 0 \iff \gcd(n, m) = 1, \quad (10.5)$$

because $d(nm) = d(n)d(m)$ exactly for coprime n, m and is strictly smaller otherwise.

For example,

$$\Delta S_{\text{mul}}(3, 4) = \ln 6 - \ln 2 - \ln 3 = 0, \quad (10.6)$$

$$\Delta S_{\text{mul}}(2, 6) = \ln 6 - \ln 2 - \ln 4 = \ln(3/4) < 0. \quad (10.7)$$

Where the framework attaches a verification-cost functional $J(n, m, \varepsilon)$ to this deficit, that functional is an additional programme definition; the sign and coprimality statements above do not depend on it.

10.3. T20: Gaussian prime splitting and divisor classes

Equation (9.6) is the prime-level divisor-class census induced by the standard splitting of rational primes in $\mathbb{Z}[i]$. The function $d_{\mathbb{Z}[i]}$ is multiplicative on coprime rational arguments, and its Dirichlet series is given by Equation (9.7).

The ideal-counting series is a different object:

$$F_{\mathbb{Q}(i)}^{\text{id}}(1, s) = \zeta_{\mathbb{Q}(i)}(s) = \zeta(s)L(\chi_{-4}, s). \quad (10.8)$$

At $s = 2$ it is approximately 1.50670, so the standard Gaussian divisor-class series is approximately 2.5 times the ideal-counting series at that point. The two towers count different equivalence classes; they should not be numerically conflated.

11. Divisor-Lattice Geometry

The preceding chapters developed the framework on the full set of natural numbers without privileging any subsequence. Yet arithmetic itself supplies a canonical filtration: the prime numbers, indexed by their ordinal position. This chapter examines what happens when the machinery is applied not to all integers but to the prime rail

$P = (p_1, p_2, p_3, \dots) = (2, 3, 5, 7, \dots)$ and to its iterated counterparts. The result is a remarkable inversion: purification of the integer set by prime extraction does not simplify the eigenform geometry but relocates it, transferring complexity from value-space to index-space.

11.1. Extraction and re-indexing

The prime rail is defined by $P(n) = p_n$, taking each natural number n to the n th prime. This is a re-indexing operation: the set of primes inherits the ordinal structure of $\mathbb{N}$, but the arithmetic properties of the values p_n bear no simple relation to those of their indices n . Every rail value is prime, so

$$H_{\text{fact}}(p_n) = \ln 2 = \lambda_{\min}.$$

All primes therefore have the same divisor-cardinality entropy. Under the angular divisor-state metric of Chapter 14, any two distinct rail values also satisfy $d_F(p_m, p_n) = 2\pi/3$.

In value-space, the prime rail is homogeneous with respect to divisor cardinality: every prime has the two-state divisor cell $C_2 \cong \mathbb{C}^2$ and fibre entropy $\ln 2$. Chapter 21 uses C_2 as an encoding of the absence/presence choice for a prime factor. It is not an irreducible S_2 representation claim.

11.2. Superprimes and iterated rails

The re-indexing operation iterates naturally. Define the superprime rail by $P^2(n) = p_{p_n}$. Its opening terms are 3, 5, 11, 17, 31, 41, 59, ...: primes whose index in the full prime sequence is itself prime. The superprime rail is a sub-rail of the prime rail; equivalently, it is the pullback of the prime-index predicate along the ordinal map. Its counting function has the familiar leading asymptotic $\pi_2(x) \sim x/(\ln x)^2$. Every element of P^2 is prime, and its index p_n is also prime, so both the value-sector and index-sector have fibre entropy $\ln 2$. In the declared two-state encoding this gives $C_2 \otimes C_2$. Higher iterates P^k are defined by repeated composition of the prime map; their counting functions have leading behaviour $\pi_k(x) \sim x/(\ln x)^k$. Whether structurally new behaviour appears for $k \geq 3$ remains open (OP-3). What is established is that the coordinate heatmaps of the prime and superprime rails display a marked regularisation compared to the full-integer heatmap: the vertical striations in the fibre-entropy row of the SOES-1000 integer coordinate heatmap collapse to near-uniform bands on the prime rail, and the contact-impedance score stabilises to a sustained high-impedance band. The superprime rail intensifies the effect: the ordinal-irreducibility row, noisy on the prime rail, becomes a clean periodic lattice on P^2 .

11.3. Prime-rail renormalisation

The deeper significance of the prime rail is methodological. When the eigenform engine is restricted to primes, the value-space degeneracy forces the observer to attend to index-space structure. The primes p_n and p_m are indistinguishable by observables that depend only on the two-element divisor-cardinality cell, yet they occupy different ordinal positions and therefore participate in different tensor products when the AEE is enriched. Purification creates an ordinal shadow: structure that existed in the value-space of the full integers relocates to the index-space of the rail. This phenomenon — the displacement

of complexity from values to indices under restriction to irreducible elements — recurs throughout the theory and motivates the enriched state construction.

11.4. The enriched prime-rail state

To capture the index-space structure that the prime rail reveals, the eigenform at a rail element must encode both value and position. Define the enriched prime-rail state by

$$\text{AEE}^+(p_n) := \text{AEE}(p_n) \otimes \text{AEE}(n) \cong C_2 \otimes \text{AEE}(n).$$

Here C_2 is the base two-state divisor cell for the prime value, while $\text{AEE}(n)$ carries the declared coordinates of the index. The enriched state does not merely record that p_n is prime; it also records the arithmetics of its ordinal index n . Its declared vector-space dimension is $2d(n)$, the product of the two-state value cell and the index-sector dimension.

11.5. Value-prime versus index-composite

The enriched construction exposes a distinction invisible to ordinary arithmetic. Consider $p_4 = 7$. The value 7 is prime, so its value-sector is C_2 and $H_{\text{fact}}(7) = \ln 2$. The index $4 = 2^2$ is composite, with the declared encoding $\text{AEE}(4) = \text{Sym}^2(C_2)$ of dimension 3 and fibre entropy $\ln 3$. Thus

$$\text{AEE}^+(p_4) = C_2 \otimes \text{Sym}^2(C_2),$$

a six-dimensional encoded state with a two-state prime value-sector and a symmetric-square index-sector. The rail element 7 is value-prime but index-composite. This distinction generates four rail-state categories:

Table 11.1.: *Classification of enriched rail states.*

Category	Value / index	Example	Entropy (value / index)	Enriched form
VP-VI	prime / prime	$p_3 = 5$	$\ln 2 / \ln 2$	$C_2 \otimes C_2$
VP-CI	prime / composite	$p_4 = 7$	$\ln 2 / > \ln 2$	$C_2 \otimes \text{AEE}(n)$
VC-VI	composite / prime	semiprime near-twin	$> \ln 2 / \ln 2$	$\text{AEE}(m) \otimes C_2$
VC-CI	composite / composite	non-rail integer	$> \ln 2 / > \ln 2$	$\text{AEE}(m) \otimes \text{AEE}(n)$

The VP-VI entries are the superprimes: every $P^2(n)$ falls in this class. The VP-CI entries are the remaining primes, whose index-composite structure imports the full diversity of the divisor lattice into an otherwise homogeneous rail. The VC-VI category is conjecturally populated by certain semiprimes adjacent to superprimes, though no systematic enumeration has been completed. The VC-CI category is the generic integer case, recovered when the rail restriction is lifted.

11.6. The ordinal-fibre product

The enriched states support a declared product that fuses the two identical prime-value cells while tensoring the index sectors. For rail elements p_m and p_n , define

$$\text{AEE}^+(p_m) \odot \text{AEE}^+(p_n) := C_2 \otimes (\text{AEE}(m) \otimes \text{AEE}(n)).$$

The symbol $\odot$ is therefore a branch-specific fusion product, not the ordinary tensor product of the two complete enriched states.

When $\gcd(m, n) = 1$, the declared coprime tensor rule identifies $\text{AEE}(m) \otimes \text{AEE}(n)$ with $\text{AEE}(mn)$, so the ordinal-fibre product has the same encoded form as $\text{AEE}^+(p_{mn})$. This gives a multiplicative law on the VP-VI sub-rail — the superprimes under index-coprimality — that mirrors the multiplicativity of the AEE on coprime integers. The structure is analogous to, though not identical with, the bialgebra structure of Chapter 14. The construction admits a representation-theoretic reading. The value-sector C2 is the lowest rung of the automorphic ladder (L_0 baseline). The index-sector $\text{AEE}(n)$ may occupy any rung depending on the arithmetic of n . The enriched state is therefore a tensor product of two ladder elements at different heights, suggesting that the enriched prime rail samples a two-parameter family of automorphic representations in a way the individual rails cannot. The superprime rail, confined to $C_2 \otimes C_2$, remains at the $L_0 \otimes L_0$ baseline — explaining its comparative regularity.

11.7. The CV theorem and the phase transition

For $N > 1$, let μ_N and σ_N be the mean and population standard deviation of the positive divisors of N . Define

$$\text{CV}(N) = \frac{\sigma_N}{\mu_N}. \quad (11.1)$$

The empirical programme uses this dimensionless spread as a predictor of spectral statistics.

T-CV1 (empirical, Class E). On the stated CIM ensemble,

$$\beta_{\text{eff}}(N) \approx 1 - e^{-a \text{CV}(N)}, \quad a \approx 0.88, \quad (11.2)$$

with reported Pearson correlation 0.905 and RMSE 0.049. The quantity β_{eff} is a fitted normalized repulsion coordinate. It is not the adjacent-gap ratio $\langle r \rangle$.

T-CV2 (proved, Class N). For squarefree $N = \prod_{i=1}^r p_i$,

$$1 + \text{CV}(N)^2 = \prod_{i=1}^r (1 + \text{CV}(p_i)^2) = \prod_{i=1}^r \left[1 + \left(\frac{p_i - 1}{p_i + 1} \right)^2 \right]. \quad (11.3)$$

In particular,

$$\text{CV}(p) = \frac{p - 1}{p + 1}, \quad (11.4)$$

so the CV of a prime is *not* constant: it equals $1/3$ only at $p = 2$ and tends to 1 as $p \rightarrow \infty$.

CVT-3 (proved, Class N). For fixed divisor count $d(N) = k$,

$$\sup_{d(N)=k} \text{CV}(N) = \sqrt{k - 1}, \quad (11.5)$$

with the supremum approached by prime powers $N = q^{k-1}$ as $q \rightarrow \infty$. For $k = 3$ the limiting value is $\sqrt{2}$.

The fitted exponent $\nu \approx 0.262$ is numerically close to $1/4$, but no shared physical universality class is established. The prime rail also needs careful interpretation: although $H_{\text{fact}}(p) = \ln 2$ is constant, the divisor CV varies with the prime value by Equation (11.4). Any additional regularity of the superprime rail must therefore be demonstrated by observables other than a constant-CV premise.

Table 11.2.: *CV results and phase-transition context.*

Item	Statement	Status	Key parameters
T-CV1	$\beta_{\text{eff}} \approx 1 - e^{-a\text{CV}}$	Empirical	$a \approx 0.88$; reported $r = 0.905$; RMSE 0.049
T-CV2	Eq. (11.3)	Proven in the framework	Squarefree N
CVT-3	$\sup_{d(N)=k} \text{CV}(N) = \sqrt{k} - 1$	Proven in the framework	Prime-power asymptote
Phase	$H_* \in (\ln 6, \ln 7)$	Empirical location	Transition lies between fibres with $d = 6$ and $d = 7$; $\nu \approx 0.262$
Continuum	$\beta_{\text{eff}} \sim \varepsilon^{0.21}$	Empirical fit	Phase-disorder branch

12. Divisor Channels as Exact Closure Classes

Frame declaration. This chapter applies to cyclic axes $\mathbb{Z}_n$ and their character/Fourier modes. The divisor-channel result is exact in that branch. It is not asserted for arbitrary graphs, open intervals, non-periodic domains, or arbitrary local couplings.

Let a cyclic axis have length n and character index $k \in \mathbb{Z}_n$. The orbit closes after the smallest positive p satisfying $pk \equiv 0 \pmod{n}$. Hence

$$p_n(k) = \frac{n}{\gcd(n, k)}.$$

Theorem 12.1 (Exact-period channel census). *The set of possible closure periods is exactly the divisor set of n :*

$$\{p_n(k) : k \in \mathbb{Z}_n\} = \text{Div}(n).$$

The number of modes of exact period $p \mid n$ is Euler's totient $\varphi(p)$.

Proof sketch. $p_n(k) = p$ iff $\gcd(n, k) = n/p$. Writing $k = (n/p)u$, exact period p requires $\gcd(u, p) = 1$, giving $\varphi(p)$ residues. ◦

Evidence status. Class K as arithmetic/group theory; independently exhaustively checked in the present programme through $n \leq 4096$. The new content is the measurement interpretation: exact periods are treated as typed closure channels.

Thus the one-dimensional channel count is

$$C_1(n) = \tau(n).$$

For a separable d -dimensional torus with side lengths $n_1, \dots, n_d$, a channel label is

$$\chi = (p_1, \dots, p_d), \quad p_i \mid n_i,$$

and the number of channel labels is

$$C_d = \prod_{i=1}^d \tau(n_i).$$

12.1. Color, multiplicity, and eigenvalue are different facts

A divisor channel is a closure class, not an eigenvalue. Its multiplicity is $\varphi(p)$. A particular operator may attach an eigenvalue λ_p to the channel, but distinct channels can share the same eigenvalue. This distinction was observed in every one of the 36 arrangement–tile combinations in the first hierarchical sweep: structural channel identity survived even when several labels collapsed to the same numeric damping level.

12.2. Arithmetic-symmetric coupling branch

Let a cyclic coupling kernel depend on a displacement r only through its exact orbit period

$$p(r) = \frac{n}{\gcd(n, r)}.$$

For example,

$$w(r) = p(r)^{-\alpha}.$$

Such a kernel is constant on gcd-orbits. Its Fourier multiplier is correspondingly constant inside the exact-period channel classes. In the programme this channel constancy held to floating-point precision for $\alpha = 1$ and $\alpha = 2$.

Ordinary nearest-neighbour diffusion does not satisfy this stronger arithmetic symmetry in general. The negative control failed channel constancy at several small n and, in the larger batch, exact divisor-channel degeneracy occurred in only one of 1,020 tested nearest-neighbour sizes. Generic random symmetric circulant kernels produced zero exact hits in 3,976 tests.

Bounds and portability. Cyclic closure gives the divisor taxonomy. Divisor-quantized damping requires additional symmetry in the coupling law. The first does not imply the second.

Failure condition. Inferring arithmetic-symmetric coupling from cyclicity alone is an overgeneralization. Equating channel labels with distinct measured frequencies is an identifiability failure when spectral degeneracies are present.

13. Arithmetic Shadows and Operator Resolution

Frame declaration. The arithmetic-shadow branch uses positive-divisor incidence. Other fibre families in the book generate different shadows. The overlap formulas here are exact only for the declared divisor frame.

Let $D(n)$ be the positive-divisor set. Then

$$D(a) \cap D(b) = D(\gcd(a, b)),$$

so the exact overlap count is

$$O(a, b) = \tau(\gcd(a, b)).$$

The source-normalized shadow is

$$\Omega(a \rightarrow b) = \frac{\tau(\gcd(a, b))}{\tau(a)}.$$

It is intentionally asymmetric: it answers what fraction of the source's divisor channels remain visible in the target.

13.1. Prime shadows

For prime p , $D(p) = \{1, p\}$, therefore

$$\Omega(p \rightarrow n) = \begin{cases} \frac{1}{2}, & p \nmid n, \\ 1, & p \mid n. \end{cases}$$

Every prime casts a permanent one-half background on the raw normalized plane because the unit divisor is universal. Multiples of p form full-overlap stripes. On a common-mode-subtracted plane that removes 1, distinct primes instead have zero overlap.

Prime powers add graded levels. For p^2 , the source has three divisor channels $\{1, p, p^2\}$ and source-normalized overlap levels $1/3$, $2/3$, and 1. Composite shadows superpose these prime-power coordinates through

$$\tau(\gcd(a, b)) = \prod_q (\min(\alpha_q, \beta_q) + 1),$$

for $a = \prod q^{\alpha_q}$ and $b = \prod q^{\beta_q}$.

13.2. Operator fingerprints

In the nonlinear period branch the action of order r on every divisor channel of a fixed modulus n depends only on $\gcd(n, r)$.

Theorem 13.1 (Operator observational equivalence). *For fixed n , two positive interaction orders r, s induce the same map on all divisor channels of n iff*

$$r \sim_n s \iff \gcd(n, r) = \gcd(n, s).$$

Proof sketch. For every $p \mid n$, $\gcd(p, r) = \gcd(p, \gcd(n, r))$. Hence the complete channel action is determined by the gcd fingerprint. Conversely the channel $p = n$ exposes $\gcd(n, r)$ directly through $\mathcal{S}_r(n) = n / \gcd(n, r)$. ◦

This motivates the resolving power of a modulus against a candidate operator set C :

$$R_C(n) = |\{\gcd(n, r) : r \in C\}|.$$

For $C = \{2, \dots, 12\}$, the high-range sweep found that $n = 2520$ is the smallest modulus giving full resolution $R_C = 11$. Its gcd fingerprints are distinct for every candidate; notably $r = 11$ occupies the unique residual fingerprint 1 because $11 \nmid 2520$ while every other candidate has a different gcd.

13.3. Divisor richness and measurement capacity

The sweep to 10^6 found divisor-count records

$$\tau(1680) = 40 \quad (n \leq 2048),$$

$$\tau(7560) = 64 \quad (n \leq 10^4),$$

$$\tau(83160) = 128 \quad (n \leq 10^5),$$

and

$$\tau(720720) = 240 \quad (n \leq 10^6).$$

Divisor richness and resolving power correlate strongly but are not identical because resolution depends on whether the available prime powers intersect the candidate operator family. In the 10^6 sweep the correlation between $\tau(n)$ and $R_C(n)$ rose from approximately 0.79 for $C = 2 : 12$ to 0.96 for $C = 2 : 128$.

13.4. Inverse identifiability as a quotient

In the 500-case inverse-order experiment, the true interaction order always lay in the minimum-error observational equivalence class, but unique identification occurred in only 8.6% of cases. This is not a defect in the estimator; it is the information boundary encoded by $r \sim_n s$. The correct output of the inverse problem is therefore an equivalence class unless the measurement modulus has sufficient resolving power.

Failure condition. Reporting a unique microscopic operator when only a gcd-equivalence class is identifiable is evidence inflation. Treating the universal 1 channel as stochastic noise is also incorrect: it is deterministic common mode in this frame.

14. Observer Bounds

This chapter separates bounds on the declared measurement instrument from claims about integers independently of that instrument. The principal objects are normalized divisor states, their angular overlap, a corrected skip-set criterion, an arithmetic uncertainty construction, and a family of readout maps.

14.1. Information-theoretic limits

For every prime p ,

$$H_{\text{fact}}(p) = \ln 2 = \lambda_{\min}. \quad (14.1)$$

The constant is simply the log-cardinality of the two-element divisor set $\{1, p\}$. In the later tensor encoding it is the logarithm of $\dim C_2 = 2$, where $C_2 \cong \mathbb{C}^2$ is a two-state factor cell.

For squarefree $n = \prod_{j=1}^r p_j$,

$$H_{\text{fact}}(n) = r \ln 2 = \omega(n) \lambda_{\min}, \quad (14.2)$$

whereas for $n = p^k$, $H_{\text{fact}}(n) = \ln(k + 1)$.

14.1.1. The angular geometry of divisor states

Attach to each positive integer the normalized divisor state

$$|\psi_N\rangle = \frac{1}{\sqrt{d(N)}} \sum_{k|N} |k\rangle. \quad (14.3)$$

Then

$$\langle \psi_{N_1} | \psi_{N_2} \rangle = \frac{d(\gcd(N_1, N_2))}{\sqrt{d(N_1)d(N_2)}}. \quad (14.4)$$

The framework defines the associated angular distance by

$$d_F(N_1, N_2) = 2 \arccos |\langle \psi_{N_1} | \psi_{N_2} \rangle|. \quad (14.5)$$

The factor of two is part of the framework's convention; the formula should therefore be read as a defined Fisher-type/angular metric rather than as an unqualified identification with every convention for the Bures or Fubini–Study distance.

Two consequences are immediate:

$$d_F(1, N) = 2 \arccos(d(N)^{-1/2}) = 2 \arccos(e^{-H_{\text{fact}}(N)/2}), \quad (14.6)$$

$$d_F(p, q) = \frac{2\pi}{3} \quad (p \neq q \text{ prime}). \quad (14.7)$$

Also,

$$d_F(p, p^2) = 2 \arccos\left(\frac{2}{\sqrt{6}}\right). \quad (14.8)$$

14.1.2. The refined skip-set theorem

Antipodality is impossible in this embedding because every positive-divisor support contains 1. For $n, m > 1$ the following statements are equivalent:

- (1) $\gcd(n, m) = 1$;

- (2) $\text{Div}(n) \cap \text{Div}(m) = \{1\}$;
- (3) the normalized overlap is minimal for the fixed pair $(d(n), d(m))$,

$$\langle \psi_n | \psi_m \rangle = \frac{1}{\sqrt{d(n)d(m)}}; \quad (14.9)$$

- (4) $\Delta S_{\text{mul}}(n, m) = 0$;
- (5) in the declared prime-exponent tensor encoding, the state of a coprime product factorises as the tensor product of the factor states.

The geometric statement is therefore *minimal overlap*, not antipodality. For example, 4 and 9 have overlap $1/3$ and $d_F(4, 9) = 2 \arccos(1/3) < \pi$.

The asymptotic density of the skip set is

$$\lim_{N \rightarrow \infty} \frac{1}{N^2} \#\{(n, m) \in [1, N]^2 : \gcd(n, m) = 1\} = \frac{6}{\pi^2}. \quad (14.10)$$

14.2. The arithmetic uncertainty construction

Let $V = d(n)$ and define the divisor-indexed Jacobi matrix with zero diagonal and off-diagonal coefficients

$$\beta_k = \sqrt{\frac{k(k+2)}{4k^2 - 1}}, \quad k = 1, \dots, V-1. \quad (14.11)$$

Define

$$B_H(V) = \left(\sum_{k=1}^{V-1} \beta_k^2 \right)^{1/2}, \quad (14.12)$$

$$\Delta H(n) = \sqrt{\text{Var}_{d|n}(\ln d(d))}, \quad (14.13)$$

$$P(n) = \Delta H(n) B_H(d(n)). \quad (14.14)$$

Here the inner $d(d)$ means the divisor-count function applied to each divisor of n ; this is what makes a prime produce the two-point set $\{0, \ln 2\}$.

Arithmetic Uncertainty Principle (Class N, within these definitions). The programme states

$$P(n) \geq \frac{\ln 2}{2}, \quad (14.15)$$

with equality exactly for primes. At a prime, $\Delta H(p) = \ln 2/2$ and $B_H(2) = 1$. The off-diagonal energy has $B_H(V) \sim \sqrt{V}/2$. The theorem is a statement about the particular observables in Equations (14.12)–(14.14); no frame-independent uncertainty constant is asserted.

14.3. The self-dual divisor/index correspondence

The divisor involution $\delta \mapsto n/\delta$ has a fixed point exactly when n is a perfect square, in which case the unique fixed divisor is $\sqrt{n}$. Thus

$$\#\text{Fix}(\delta \mapsto n/\delta) = \mathbf{1}_{\text{sq}}(n). \quad (14.16)$$

For the alternating chiral grading on the divisor-indexed Jacobi chain, the index is 1 for odd $d(n)$ and 0 for even $d(n)$; since $d(n)$ is odd exactly for squares, the three indicators coincide in this branch. The “Majorana-type” terminology is an analogy to a protected self-conjugate zero mode and is not a physical-particle claim. Statements about the genus of a programme-specific prime-power spectral curve require the separate spectral-curve construction and are not used as part of the square-detection theorem.

14.4. Readout, attenuation, and irreversible projection

Consider

$$P_\varepsilon(J) = \text{diag}(J) + (1 - \varepsilon) \text{off}(J), \quad 0 \leq \varepsilon \leq 1. \quad (14.17)$$

A crucial distinction is algebraic. For $0 \leq \varepsilon < 1$, P_ε is invertible in exact arithmetic: the off-diagonal part can be recovered by division by $1 - \varepsilon$. Only $P_1(J) = \text{diag}(J)$ is a non-invertible projection and therefore necessarily destroys off-diagonal information.

For a reversible flow Φ_t , define the round-trip residual

$$R_{\text{rt}}(\varepsilon) = \|J_0 - \Phi_{-T}(P_\varepsilon(\Phi_T(J_0)))\|_F. \quad (14.18)$$

The reported programme correlation between this residual and an independent attenuation/loss diagnostic is an empirical result. It must not be read as a proof that every $\varepsilon > 0$ is intrinsically irreversible. In the exact map above, irreversibility enters at the singular endpoint $\varepsilon = 1$ or through any additional quantisation, noise, thresholding, or non-invertible measurement operation explicitly added to the branch.

14.4.1. What overlap resolves

Marginal probabilities of the state in Equation (14.3) are uniform and reveal only $d(N)$. Pairwise overlap adds shared-divisor information through Equation (14.4). In particular, its numerator is $d(\gcd(a, b))$; the overlap therefore yields a *gcd-derived divisor-count observable*, not the gcd value itself. Since many integers have the same divisor count, unique recovery of the microscopic gcd generally requires additional observables.

15. Prime Saturation and Slack

The arithmetic of coprimality acquires geometric meaning through the skip-set construction of Chapter 14. This chapter records two empirical spectral programmes: a phase-disorder continuum and gap-based diagnostics. Their statistics are kept distinct from the theorem-level arithmetic results.

15.1. The phase continuum

For a partial-phase ensemble with disorder parameter $\varepsilon \in [0, 1]$, the programme fits a normalized repulsion coordinate

$$\beta_{\text{eff}} \sim \varepsilon^{0.21}. \quad (15.1)$$

Here β_{eff} is an internally fitted coordinate scaled so that the Poisson-like endpoint is near 0 and the strongly repulsive endpoint is near 1. It is not the adjacent-gap ratio.

Where level-spacing ratios are reported, the notation is instead $\langle r \rangle$. Standard benchmarks are approximately

$$\langle r \rangle_{\text{Poisson}} \approx 0.386, \quad \langle r \rangle_{\text{GOE}} \approx 0.536, \quad \langle r \rangle_{\text{GUE}} \approx 0.603. \quad (15.2)$$

Consequently a fitted value such as $\beta_{\text{eff}} \approx 0.58$ must not be numerically compared to the GOE/GUE values in Equation (15.2). The two observables answer different questions.

15.2. The stochastic Toda mechanism

The synthetic stochastic branch uses

$$dL = (1 - \varepsilon)[B, L]dt + \varepsilon \Xi(t) dW_t. \quad (15.3)$$

At $\varepsilon = 0$ this reduces to deterministic isospectral Toda flow; the noise term breaks exact isospectrality when $\varepsilon > 0$. The programme reports a transition near $\varepsilon_c \approx 0.2$ and GUE-like adjacent-gap statistics above that regime, with $\langle r \rangle$ reported in the approximately 0.58–0.60 range. This is an empirical classification, not an identity with β_{eff} .

The reported holonomy diagnostic

$$B_H = \|L - \text{diag } L\|_F \quad (15.4)$$

was fitted by a log-normal stationary distribution above the stated critical regime, with reported log-scale width $\sigma_B \approx 0.67$.

15.3. Spectral gaps and crisis events

Let

$$\delta_{\min} = \min_i |\lambda_{i+1} - \lambda_i| \quad (15.5)$$

be the minimum ordered spectral gap. The programme treats narrowing gaps as a regime diagnostic. The retained numerical record gives gaps 0.4439 at $n = 6$ and 0.3230 at $n = 8$. These two values alone do not establish a scaling law; no additional numerical datum is inferred.

A decay law of the form

$$B_H(t) = O(k^3 n^3 e^{-\delta_{\min} t}) \quad (15.6)$$

is the programme's gap-controlled model. Equation (15.6) should be treated as branch-specific until its derivation and constants are established independently.

The theta-oracle proposal uses a hyperelliptic spectral curve written in the standard form

$$\mu^2 = \prod_{i=1}^{V-1} (\lambda - e_i). \quad (15.7)$$

For a degeneration $e_{i+1} - e_i \rightarrow 0$, the programme proposes the *conjectural* scaling

$$t_{\text{collapse}} \asymp \delta_{\min}^{-1} \frac{\vartheta(z_0; \tau)}{\vartheta(z_*; \tau)}. \quad (15.8)$$

No algebraic proof of Equation (15.8) is claimed. Its role is to define a falsifiable target for the numerical programme, not to certify a closed-form collapse law.

16. The Measurement Axis

The framework separates reversible transport from non-invertible readout. That separation is algebraic: a map is not irreversible merely because it attenuates coordinates; irreversibility requires loss of invertibility or an explicitly lossy stochastic/quantised operation.

16.1. The reversible layer: dynamics

For a real symmetric Jacobi matrix J , write J_+ for its strictly upper triangular part and $J_- = J_+^T$. The finite non-periodic Toda Lax flow is

$$\dot{J} = [B(J), J], \quad B(J) = J_+ - J_-.$$
 (16.1)

Since $B(J)$ is skew-symmetric, the solution has the orthogonal-similarity form

$$J(t) = Q(t)^T J(0) Q(t),$$
 (16.2)

for the corresponding orthogonal propagator $Q(t)$, and therefore preserves the spectrum and trace invariants.

Autonomy implies local time reversibility of the exact flow map whenever the solution exists uniquely, but it does not by itself imply quasi-periodicity or a prime/composite classification of Toda trajectories.

For a traceless real-symmetric 2×2 Jacobi slice $J = a\sigma_3 + b\sigma_1$, fixed spectrum gives $a^2 + b^2 = \text{const}$, which is a circle. The full $SU(2)$ coadjoint orbit in $\mathfrak{su}(2)^*$ is a two-sphere only after the third Pauli direction is included. The real-symmetric Jacobi family is a slice of that orbit, not the whole sphere.

16.2. The irreversible layer: projection and readout

The readout family P_ε is defined in Equation (14.17). For $\varepsilon < 1$ it is a reversible linear attenuation in exact arithmetic. At $\varepsilon = 1$ it becomes the projection

$$P_1(J) = \text{diag}(J),$$
 (16.3)

which retains the diagonal Cartan subalgebra and discards the off-diagonal component. This is a Lie-algebra projection, not literally a group map $SU(2) \rightarrow U(1)$.

The round-trip residual is given by Equation (14.18). A nonzero residual after an invertible P_ε can measure attenuation relative to a pipeline that does not apply the inverse; it is not by itself proof of information-theoretic destruction. Genuine irreversible readout occurs when the measurement operation is non-invertible, as at $\varepsilon = 1$, or when an additional lossy operation is declared.

16.3. The skip set as the zero-loss locus

The skip set

$$S = \{(n, m) : \gcd(n, m) = 1\}$$
 (16.4)

is exactly the zero set of multiplicative entropy production:

$$\Delta S_{\text{mul}}(n, m) = 0 \iff (n, m) \in S.$$
 (16.5)

Chapter 14 gives four aligned readings: no nontrivial shared prime factors; only the unit divisor in the support intersection; minimum normalized divisor-state overlap for fixed divisor counts; and exact factorisation in the declared prime-exponent tensor encoding. Its classical asymptotic density is Equation (14.10).

16.4. Failure modes

The time-crystal interpretation of the round-trip residual remains recorded as falsified. In the reported driven tests, no stable subharmonic period-doubling regime was observed across the stated configurations. The safe interpretation is narrower: residuals diagnose disagreement between forward transport, declared readout, and reverse transport. Whether that disagreement represents reversible attenuation, numerical error, or genuine information loss depends on the algebraic properties of the readout map and must be stated branch by branch.

The same discipline applies to physical language. Toda isospectrality is a classical integrable-system property; the Cartan projection is an algebraic operation; and the divisor-state phase is a coordinate of the constructed state. None of these facts alone establishes a physical quantum system.

17. Computational Morphospace

The resemblance to quantum computing was not the destination. It was the first shadow of a deeper structure.

If quantum computing, Boolean logic, and spectral-arithmetic computation share a common silhouette, the shared shape is not quantum mechanics. It is a deeper regularity: stable computation arises when a substrate is constrained by operators, governed by correction protocols, and certified through measurement semantics. This chapter makes that regularity explicit — constructing the space in which computational species live, defining what makes a species irreducible, and locating Eigenalgebra Arithmetics within that space as a candidate whose prime status remains open. The chapter closes Part II by introducing the operator-constraint morphospace and the concept of computational primes. Both are formalisations: coherent, operationally defined, and productive of testable predictions, but not yet resting on theorem-grade foundations.

17.1. The eight-axis typed space

A computational system is not described by its substrate alone. A Turing machine is not merely a tape; a quantum circuit is not merely a Hilbert space; a spectral-arithmetic engine is not merely a divisor lattice. What distinguishes a computational species — a stable, recognisable type of computation — is coordinated structure across multiple independent axes. The operator-constraint morphospace, denoted K , is the typed eightaxis space:

$K = (S, C, O, I, R, M, G, E)$ S is the substrate or state space. C is the constraint system: rules delimiting admissible states and quantities fixed under evolution — unitarity and hermiticity in quantum computing, logical consistency and fan-out limits in Boolean computation, constraint cells and isospectral Toda invariants in Eigenalgebra. O is the set of admissible operators.

I is the interaction or composition law: the tensor product, circuit concatenation, or the coprime product $\psi(ab) = \psi(a) \otimes \psi(b)$. R is the resource theory: what expands the easy closed subtheory into the universal — entanglement plus magic states for quantum computing, memory plus universality for Boolean computation, chromodynamic eigenflow curvature for Eigenalgebra. M is the measurement or readout semantics. G is governance: correction, projection, and control. E is the evidence logic: the rules by which claims are certified or rejected. The morphospace is typed: these axes are not commensurable. One cannot add a substrate dimension to a governance dimension. The space is a category-theoretic product of distinct domains, and coordinates along different axes transform under different groups. This typing prevents the reduction of computational identity to any single axis.

17.2. Computational species as stable patterns

A computational species is a stable pattern across the eight axes of K . Stability is dynamical, not static. A species persists not because its coordinates are fixed, but because perturbations along any axis are compensated by responses from the others. The Spectral Observer Principle — proved as a six-theorem chain (Chapter 22) — provides one model: spectral encoding under Lanczos tridiagonalization, isospectral Toda evolution, levelcrossing detection, secular-oracle readout, and theta-function completion form a closed loop in which information is conserved, transformed, and recovered. This coherence admits degrees. Boolean computation has exceptionally strong closure on substrate, constraints, and composition, but relatively weak resource theory — its universality is trivial once memory is granted. The quantum species has rich resource theory but pays a corresponding cost in measurement semantics and governance (decoherence, threshold theorems). Eigenalgebra occupies a region where resource

theory and measurement semantics are strong — fibre entropy, Fisher distance, and contact energy are native measurements — but governance and evidence logic remain under construction.

17.3. Three species compared

Three computational species across the eight morphospace axes.

Axis	Quantum computing	Boolean computation	Eigenalgebra Arithmetics
S – substrate	Hilbert space; qubit tensor products	Discrete $\{0, 1\}^n$; finite-state machines	Spectral-arithmetic lattice; divisor fibres $F_n = \pi^{-1}(n)$; Jacobi state space
C – constraints	Unitarity, Hermiticity, no-cloning	Logical consistency; finite memory	Constraint cells $C = (X, E, \pi, D)$; Toda invariants $\text{tr}(L^k)$; floor $\lambda_{\min} = \ln 2$
O – operators	Unitaries, projections, CPTP maps, Clifford+T	Boolean gates (NAND, NOT), Toffoli	Jacobi operators $J(n)$, Toda flow, chiral Dirac D_n , Cartan projection P_i
I – composition	Tensor product; entangling gates	Circuit concatenation; parallel composition	Coprime product $\psi(ab) = \psi(a) \otimes \psi(b)$; bialgebra Δ
R – resource	Entanglement and magic-state resources	Memory and functional completeness	Chromodynamic eigenflow curvature; fibre-entropy gradient; divisor-lattice completeness
M – measurement	Projective measurement; Born rule	Deterministic bit readout	Divisor-state angular distance, contact energy, spectral gap, uncertainty product
G – governance	QEC thresholds; fault-tolerant design; decoherence-free subspaces	Classical error-correcting codes	Conjugate Substrate Invariance modes; Reciprocity Law; brake/projection governance
E – evidence	Bell/complexity/threshold theorems	Completeness proofs	Five-tier evidence system; promotion gates; null-control discipline

Several patterns emerge. Quantum computing and Eigenalgebra share density across the measurement axis: both have native measurement formalisms (Born-rule probabilities versus the divisor-state angular metric), native uncertainty principles (Heisenberg versus arithmetic uncertainty), and native spectral readouts. Boolean computation lacks this richness; its measurement semantics are trivial by design, which is both a strength (determinism) and a limitation (no internal metrology). Boolean computation and Eigenalgebra share a deterministic substrate: both evolve states through rules containing no irreducible randomness. This distinguishes them from quantum computing, where measurement outcomes are probabilistic in a structurally central manner. Eigenalgebra reproduces structural features of quantum formalism — spectral operators, chiral symmetry, information-theoretic bounds, phase structure — without invoking Hilbert-space superposition or physical entanglement. The Toda lattice provides deterministic isospectral flow; the Lanczos algorithm is a classical numerical method. The resemblance is real but bounded.

17.4. Computational primes

A computational prime is an irreducible stable attractor in K : a computational species that cannot be factored into simpler species, under a chosen factorisation operation, without loss of its defining computational power. Primeness is not absolute; it is relative to the class of operations admitted as decompositions. A species may be prime under circuit decomposition yet composite under oracle reduction. The analogy to arithmetic primeness is deliberate but bounded. Primes are irreducible under multiplication; computational primes are irreducible under composition of species. Just as $6 = 2 \times 3$ loses the distinct factor structures of 2 and 3 when viewed merely as "6", decomposing a computational prime into substrate-plus-constraints-plus-operators loses the coherence that makes the species powerful. The whole exceeds the sum of its axes because the axes are typed and their coherence is the species. Plausibly prime candidates. Boolean logic is plausibly a computational prime. Its substrate is minimal ($\{0, 1\}$), its constraints are complete, its operator set is functionally complete (NAND suffices), and its resource theory reduces to memory. To decompose it further — into a perceptron ensemble or reversible gate network — is to lose either universality or simplicity. Quantum computing is plausibly a computational prime. Its substrate, constraints, and resource theory form a tightly coupled package; removing any axis — eliminating superposition, replacing unitary evolution with stochastic flow, stripping entanglement — destroys the associated computational claims. The threshold theorem holds only when the full

coordinate profile is intact. Eigenalgebra Arithmetics is a candidate computational prime. It has not yet demonstrated the closure, invariant preservation, resource separation, readout fidelity, correction strength, compositionality, irreducibility, and empirical basin stability required for certification. The claim at this stage is architectural: the eight-axis profile exhibits sufficient internal coherence and sufficient differentiation from Boolean and quantum species to motivate the prime hypothesis (OP-10).

17.5. Promotion criteria

Promotion of a candidate species to a certified computational prime requires seven separate demonstrations:

1. **Closure.** The declared operator family must keep admissible states inside the species. The Jacobi/Toda branch supplies a scoped example of structural closure; arithmetic coprime composition supplies another.
2. **Invariant preservation.** The relevant invariant family must be stated explicitly and shown to survive the declared transport. Examples include spectral trace invariants $\text{tr}(L^k)$ under exact Lax flow and the square indicator in the divisor/index branch.
3. **Resource separation.** A candidate resource must be separated from an easier closed subtheory by a theorem or discriminating experiment. Chromodynamic eigenflow curvature is presently a candidate, not a certified resource monotone.
4. **Readout.** Internal states require observables with declared semantics and error budgets. Candidate readouts include the divisor-state angular metric, contact energy, spectral gaps, and the uncertainty product. The family P_ε is invertible for $\varepsilon < 1$ and becomes a non-invertible diagonal projection only at $\varepsilon = 1$.
5. **Correction.** Failure modes require operational containment and recovery rules rather than names alone. The current distortion taxonomy is a design scaffold until such rules are demonstrated.
6. **Compositionality.** Arithmetic coprime composition is defined in its own branch; extension to full morphospace composition while preserving all eight axes remains open.
7. **Empirical basin stability.** Stability must be demonstrated under predeclared perturbations across the relevant axes. Local numerical evidence does not yet establish a global basin.

17.6. Evidence-led promotion as certification mechanism

The promotion of the Spectral Observer-Eigenflow State from candidate species to certified inhabitant of K , and of Eigenalgebra from candidate computational prime to recognized prime, is governed by evidence-led promotion. Promotion decisions — whether an operator enters the admissible set, whether a claim advances tier, whether a species is recognized as prime — are driven by completed evidence gates rather than intuition, elegance, or authorial confidence. The pipeline has six stages: intuition; mathematical forging; rigor audit; load-bearing extraction; extension stress-test; and iteration, with failures feeding back to refine the candidate. For Eigenalgebra as a species, the evidence status is mixed. The corpus contains internally identified exact results, numerical verifications, empirical laws, and open claims across the substrate, constraint, and measurement axes; their local status is stated where each result is used. The resource and governance axes remain predominantly conjectural or design-stage. The evidence axis itself is operational — the five-tier system is implemented and functioning — but its effectiveness is tested only by the claims already processed. The question that closes Part II is whether a system whose evidence logic is self-referentially applied to its own morphospace coordinates can achieve stable recognition as a computational species. The answer is provisional. The SOES is a candidate.

Eigenalgebra is a candidate prime. The morphospace is a candidate architecture. What sustains the enterprise is not the certainty of these promotions but the discipline of the process by which they are sought. The measurement frame does not guarantee that the object will pass; it guarantees only that the criteria are declared in advance, applied uniformly, and revised when they fail. Part III turns from the global architecture of computational species to the local machinery: the representation-theoretic towers, the Jacobi chassis that makes spectral observation computable, and the circuit formalism that

compiles it. The morphospace tells us where a species lives. The Jacobi chassis tells us how an object moves inside it.

Part III

Self-Interaction, Operators, and Structural Dynamics

Interaction, transport, and measurement are separated before they are recomposed.

18. The Self-Interaction Calculus

Frame declaration. This chapter concerns typed interaction operators. It does not assert that multiplication, attention, recurrence, reflection, covariance, and metacognition are the same operation. It studies a common construction: restricting a declared interaction law to the diagonal.

The initiating definition is deliberately small. Let X be a state space, let $\Delta_X : X \rightarrow X \times X$ be the diagonal map $\Delta_X(x) = (x, x)$, and let $\mu : X \times X \rightarrow Y$ be a typed interaction operator. Define

$$S_\mu := \mu \circ \Delta_X, \quad S_\mu(x) = \mu(x, x).$$

The square x^2 is therefore one branch, not the definition. When $\mu(a, b) = ab$ in an algebra, $S_\mu(x) = x^2$. When $\mu(x, y) = \langle x, y \rangle$, $S_\mu(x) = \|x\|^2$. When $\mu(X, Y) = XY^\top$, diagonal restriction produces a Gram-type relation object. The construction survives substrate changes only when the relevant typing survives.

18.1. Closed and relational branches

Two branches must be distinguished from the outset.

Closed self-interaction has $\mu : X \times X \rightarrow X$. Then $S_\mu : X \rightarrow X$ and the operation may be iterated:

$$x_{t+1} = S_\mu(x_t).$$

This defines genuine dynamics on the same state space.

Relational self-interaction has $\mu : X \times X \rightarrow R$. Then $S_\mu : X \rightarrow R$ constructs a relation object. To return to the state space one needs a second typed operation, for example

$$X \xrightarrow{S_\mu} R \xrightarrow{\rho} X'.$$

Self-attention is an instance of this second architecture, but the architecture is more general than attention.

18.2. Higher-order self-interaction

For an n -ary interaction $\mu_n : X^n \rightarrow Y$, define the n -fold diagonal $\Delta_n(x) = (x, \dots, x)$ and

$$S_{\mu_n}^{(n)} = \mu_n \circ \Delta_n.$$

The notation $x_\mu^{(n)}$ is reserved for interaction order, not numerical exponentiation. This prevents a recurrent error: treating “second order” as though it necessarily meant “squared magnitude.”

18.3. Diagonal observability

The diagonal is informative, but not omniscient.

Theorem 18.1 (Symmetric bilinear reconstruction). *Let $B : X \times X \rightarrow \mathbb{R}$ be a symmetric bilinear form over a real vector space and define $Q(x) = B(x, x)$. Then*

$$B(x, y) = \frac{1}{2} (Q(x + y) - Q(x) - Q(y)).$$

Thus the diagonal determines the full symmetric bilinear law.

Evidence status. Class K: polarization is classical. Its role here is architectural: it identifies when self-interaction measurements are sufficient for reconstruction.

For a general bilinear law μ , decompose

$$\mu = \mu_s + \mu_a,$$

where $\mu_s(x, y) = \frac{1}{2}(\mu(x, y) + \mu(y, x))$ and $\mu_a(x, y) = \frac{1}{2}(\mu(x, y) - \mu(y, x))$. Then

$$\mu_a(x, x) = 0,$$

so

$$S_\mu = S_{\mu_s}.$$

The antisymmetric component is real structure lying in the kernel of the diagonal observer.

Definition 18.2 (Diagonal observational equivalence). *Two interaction laws are diagonally equivalent when*

$$\mu \sim_\Delta \nu \iff \mu(x, x) = \nu(x, x) \text{ for all } x.$$

The observer identifies an equivalence class $[\mu]_\Delta$, not necessarily the microscopic operator.

This generalizes a principle already present in the observer-bound discussion: structure invisible to the chosen readout cannot be promoted to nonexistence merely because the readout cannot see it.

18.4. Differential probes

If $S(x) = \mu(x, x)$ is differentiable,

$$DS_x[h] = D_1\mu(x, x)[h] + D_2\mu(x, x)[h].$$

For bilinear μ ,

$$DS_x[h] = \mu(h, x) + \mu(x, h).$$

Again the first differential probe exposes the symmetrized cross-interaction, not the antisymmetric direction. Ordered cross-probes are required for that:

$$\mu_a(x, y) = \frac{1}{2}(\mu(x, y) - \mu(y, x)).$$

18.5. Interaction, recurrence, and self-reference

Three axes are kept distinct:

$$\text{self-interaction: } S(x) = \mu(x, x),$$

$$\text{recurrence: } x_{t+1} = F(x_t),$$

$$\text{self-reference: } F(F) \text{ or an encoding thereof.}$$

They may be composed, for example

$$x_{t+1} = F(x_t, S_\mu(x_t)),$$

but none implies the others.

Branch contract. The self-interaction calculus is the general layer. Squaring, Gram formation, attention-like relation lifting, divisor-channel power maps, and reflective controllers are branches. A branch is admitted only after its domain and codomain are declared.

Failure condition. Overloading x^2 across incompatible types is a typing failure. Treating diagonal observability as full operator observability is an inverse-problem failure. Calling recurrence “self-interaction” without an interaction operator is a category error.

19. Nonlinear Channel Dynamics and Modular Intermodulation

Frame declaration. The exact transition law in this chapter applies to pointwise monomial self-interaction of Fourier characters on a cyclic domain. More general nonlinearities decompose into sums of such interactions only when their representation supports that expansion.

For the pure character

$$x_j = e^{2\pi i k j / n},$$

order- r pointwise self-interaction gives

$$x_j^r = e^{2\pi i (rk) j / n},$$

so the carrier index transforms by multiplication:

$$k \mapsto rk \pmod{n}.$$

Theorem 19.1 (Divisor-channel transition). *If a pure mode has exact period p , then order- r pointwise self-interaction has exact period*

$$\mathcal{S}_r(p) = \frac{p}{\gcd(p, r)}.$$

Repeated application yields

$$p_t = \frac{p_0}{\gcd(p_0, r^t)}.$$

For $r = 2$, each iteration removes one available factor of two until the odd part remains. For general r , repeated interaction removes precisely the prime-power content of p_0 supported on primes dividing r .

Theorem 19.2 (Semigroup action). *The channel maps satisfy*

$$\mathcal{S}_r \circ \mathcal{S}_s = \mathcal{S}_{rs}.$$

Proof sketch. For each prime q , the exponent of q removed from p by $\mathcal{S}_r$ is the minimum of its exponent in p and in r . Sequential removal by s and r equals removal by the product rs .

This turns interaction order into a commutative semigroup action on the divisor lattice. The algebraic statement is exact; the interpretation of the resulting map as a physically realized transition belongs to whichever branch supplies the dynamics.

19.1. Mixed modes and modular additive closure

Let K be the set of active carrier indices. An order- r pointwise polynomial product can only generate modular sums of r input carriers:

$$K_{out} \subseteq K^{\oplus r} = \{q_1 + \dots + q_r : q_i \in K\} \pmod{n}.$$

For real fields, K includes the conjugate partners $\pm k$. The programme tested two-, three-, and four-carrier fields through orders two to four; observed support remained inside the predicted modular closure. A preregistered batch initially failed an excessively strict leakage scalar at approximately machine epsilon; a direct forbidden-bin energy diagnostic measured leakage below 5.5×10^{-28} . Both records are retained: the criterion failed, the later diagnostic explains the numerical cause.

19.2. Aliasing is part of the law

Unconstrained polynomial combination counts overstate observed support because modular sums collide. The correct prediction is the cardinality of the modular closure, not the number of ordered interaction tuples. Twenty deliberately collision-heavy tests in Major Batch 3 produced no support outside the modular prediction.

19.3. Higher dimensions

On a product torus the pure-mode rule acts coordinatewise:

$$(p_1, \dots, p_d) \mapsto \left(\frac{p_1}{\gcd(p_1, r)}, \dots, \frac{p_d}{\gcd(p_d, r)} \right).$$

The programme verified the rule in explicit 2D fields up to more than one million sites in one batch and in 3D fields up to 368,640 sites in another.

Failure condition. Calling p a temporal subharmonic without a driven-time interpretation is forbidden. Here p is an orbit/index closure period. This does not revive the falsified time-crystal claim recorded in the ledger, which concerned temporal response to periodic driving and found no period-doubling regime.

20. Cross-Scale Coupling, Tiled Sectors, and Compatibility

Frame declaration. This chapter concerns hierarchical cyclic scales and block/tile operators. Cross-scale “resonance” means compatibility of exact-period labels under a declared coupling, not physical resonance unless a physical branch supplies that interpretation.

Let an arrangement have cyclic side length A and an internal tile have cyclic side length T . The periods available on both scales are

$$\text{Div}(A) \cap \text{Div}(T).$$

By the elementary divisor identity,

$$\text{Div}(A) \cap \text{Div}(T) = \text{Div}(\gcd(A, T)).$$

This is the exact overlap set for closure channels.

In two spatial axes, an A^2 arrangement of T^2 tiles carries

$$\tau(A)^2 \tau(T)^2$$

separable structural channel labels. The 36-case arrangement–tile sweep matched this count exactly.

20.1. Weakly broken selection rules

In a synthetic channel-coupling model where only equal period labels are allowed to transfer, forbidden channels receive zero energy at zero perturbation. When a generic forbidden coupling of strength ε is introduced, first-order perturbation predicts

$$A_{\text{forbidden}} = O(\varepsilon), \quad E_{\text{forbidden}} = O(\varepsilon^2).$$

Across the implemented cross-scale sweeps the measured log–log exponents were 1.0000 and 2.0000. This is a branch-specific empirical law of the linear coupling model, not a theorem for every hierarchical system.

20.2. The tiled 6-by-6 construction

The first tiled experiment used a 4×4 arrangement of 6×6 matrix blocks. Let $M = M_s + M_a$ with $M_s^\top = M_s$ and $M_a^\top = -M_a$. Define

$$B_0 = J_4 \otimes M_s,$$

and

$$B_1 = J_4 \otimes M_s + L_4 \otimes M_a,$$

where J_4 is the all-ones matrix and L_4 the cycle Laplacian.

The plain repetition B_0 has rank six despite being 24×24 ; the repeated tiles occupy one uniform tile-space sector. The enriched B_1 has the same symmetric/diagonal self-interaction observable to

machine precision while reaching full rank 24. In the simultaneous tile-space eigenbasis it decomposes as

$$4M_s \oplus 2M_a \oplus 2M_a \oplus 4M_a.$$

Thus exact observable overlap may conceal substantial directional operator structure.

20.3. Compatibility as commutation

When tile-space operators commute, their eigensectors can be jointly resolved. Major Batch 3 extended this to several tile sizes and obtained maximum sector prediction error below 4×10^{-16} . A deliberately noncommuting perturbation mixed the frozen sectors with amplitude proportional to perturbation strength, measured exponent 1.0000.

This yields a useful compatibility criterion:

$$[C, L] = 0 \quad \Rightarrow \quad \text{common sector decomposition is admissible,}$$

whereas nonzero commutator norm quantifies a source of perspectival mixing.

Bounds and portability. The sector decomposition is exact for the constructed Kronecker/commuting branch. It is not automatically available for arbitrary tilings. The correct question is whether the transport and measurement operators commute or admit a controlled joint decomposition.

21. Tensor Encoding, CIM Symmetry, and the Two Fibre Towers

Let $K_{n,\dots,n}$ be the balanced complete k -partite graph with k colour classes of size n . A Coloured Integer Matrix (CIM) is a $kn \times kn$ integer matrix with zero diagonal blocks and arbitrary declared off-diagonal blocks.

21.1. The wreath-product symmetry

The graph automorphism group is

$$G_c \cong S_n \wr S_k, \quad (21.1)$$

with order $(n!)^k k!$: S_k permutes the colour classes and the k copies of S_n permute vertices within each class. This is an automorphism-group statement about the balanced complete multipartite graph and its block pattern.

Standard Schur–Weyl duality is different: on $V^{\otimes k}$ it identifies the commuting actions of $GL(V)$ (or $U(V)$) and the symmetric group S_k . The wreath-product symmetry in Equation (21.1) is a graph-automorphism statement, not a Schur–Weyl centralizer theorem. Wreath-product representation theory may still be useful for CIM symmetries, but that is a separate construction requiring its own proof.

21.2. The divisor tensor encoding

For the arithmetic encoding define a two-state factor cell

$$C_2 \cong \mathbb{C}^2, \quad (21.2)$$

whose basis records absence or presence of a prime factor in a divisor. If $n = \prod_{j=1}^r p_j$ is squarefree, its divisor set is the Boolean cube $\{0, 1\}^r$, so the canonical vector-space encoding is

$$\mathcal{V}(n) = C_2^{\otimes r}, \quad \dim \mathcal{V}(n) = 2^r = d(n), \quad (21.3)$$

and hence

$$H_{\text{fact}}(n) = \ln \dim \mathcal{V}(n) = r \ln 2. \quad (21.4)$$

This is a dimension identity for the declared divisor-state vector space. It does not require, and should not be described as, a non-trivial irreducible representation of S_2 ; both irreducible complex representations of S_2 are one-dimensional.

For a prime power p^a , the exponent can take the $a + 1$ values $0, \dots, a$. One convenient encoding is

$$\mathcal{V}(p^a) = \text{Sym}^a(C_2), \quad \dim \text{Sym}^a(C_2) = a + 1. \quad (21.5)$$

Thus for $n = \prod_p p^{a_p}$,

$$\mathcal{V}(n) = \bigotimes_p \text{Sym}^{a_p}(C_2), \quad \dim \mathcal{V}(n) = \prod_p (a_p + 1) = d(n). \quad (21.6)$$

Equations (21.3)–(21.6) are an exact encoding of the divisor-count identity. Any stronger representation-theoretic claim must specify the acting group and its action.

21.3. The automorphic ladder

The fibre-cardinality family

$$F(r, s) = \sum_{n \geq 1} d(n)^r n^{-s} \quad (21.7)$$

satisfies

$$\partial_r F(r, s)|_{r=0} = \sum_{n \geq 1} H_{\text{fact}}(n) n^{-s} =: \zeta_H(s), \quad (21.8)$$

where differentiation is justified in the region of absolute convergence. Likewise a character-twisted family satisfies $F_\chi(0, s) = L(\chi, s)$ and its r -derivative produces the corresponding entropy-weighted series.

Table 21.1.: Automorphic-ladder programme.

Rung	Series	Fibre	Status
L_0	$\zeta(s) = F(0, s)$	Trivial	Classical baseline
L_1	$\zeta_H(s) = \partial_r F(r, s) _0$	Factorisation	Exact derivative identity
L_χ	$\zeta_{H, \chi}(s)$	Character-twisted	Exact where the twisted series converges
$L_{\mathbb{Z}[i]}$	Gaussian fibre series	Gaussian	Defined by the selected Gaussian fibre
L_∞	—	—	Open target (OP-1)

The operator $\partial_r|_{r=0}$ is an ordinary derivation and therefore obeys the Leibniz rule. That fact alone does not define a differential graded algebra: no grading and no nilpotent differential $d^2 = 0$ are supplied here. The stronger DGA/cohomology language is therefore removed from the theorem layer.

21.4. The two fibre towers

Two natural counting problems over $\mathbb{Z}[i]$ must be kept separate. The standard Gaussian *element-divisor* count up to associates is the function used in Equation (9.7):

$$F_{\mathbb{Q}(i)}^{\text{div}}(1, s) = \zeta(s)^2 (1 + 2^{-s}) \prod_{p \equiv 1 (4)} \frac{1 + p^{-s}}{1 - p^{-s}}. \quad (21.9)$$

The *ideal-counting* tower is

$$F_{\mathbb{Q}(i)}^{\text{id}}(1, s) = \zeta_{\mathbb{Q}(i)}(s) = \zeta(s) L(\chi_{-4}, s). \quad (21.10)$$

At $s = 2$,

$$F_{\mathbb{Q}(i)}^{\text{div}}(1, 2) \approx 3.76676, \quad F_{\mathbb{Q}(i)}^{\text{id}}(1, 2) \approx 1.50670, \quad (21.11)$$

a ratio of 2.5 (up to the displayed numerical precision).

The ramified 2-adic factor $(1 + 2^{-s})$ is required by the stated definition that counts all Gaussian divisor classes of a rational integer.

More generally, ideal-counting over a quadratic field K is encoded by its Dedekind zeta function, while an element-divisor tower depends on how rational integers factor in $\mathcal{O}_K$ and need not collapse to the same L -function. Expressing the remaining element-divisor correction factors in standard automorphic terms remains an open programme rather than a theorem.

22. The Jacobi Chassis

The Jacobi matrix is the load-bearing numerical object of the spectral branch: a real symmetric tridiagonal matrix whose recurrence coefficients encode a measure and whose eigenpairs generate Gaussian quadrature. The chapter keeps classical facts, programme-specific coefficient choices, empirical claims, and conjectural prediction mechanisms explicitly separated.

22.1. Streaming Jacobi operators

For monic orthogonal polynomials the three-term recurrence is written

$$p_{k+1}(x) = (x - \alpha_k)p_k(x) - \beta_k^2 p_{k-1}(x). \quad (22.1)$$

The associated $n \times n$ Jacobi matrix has diagonal $\alpha_0, \dots, \alpha_{n-1}$ and off-diagonal $\beta_1, \dots, \beta_{n-1}$.

The programme-specific divisor chassis sets the diagonal to zero and uses

$$\beta_k = \sqrt{\frac{k(k+2)}{4k^2 - 1}}, \quad k = 1, \dots, V-1, \quad (22.2)$$

with $V = d(n)$. Equation (22.2) is a model choice. The statement that appending one moment leaves all previous recurrence coefficients untouched is valid only when one is extending a fixed underlying Jacobi operator/measure; it is not a general property of arbitrary streaming re-estimation.

22.1.1. Eigenvalues as quadrature nodes

For a Jacobi matrix associated with a positive measure μ , the eigenvalues λ_j are the Gaussian quadrature nodes and the squared first components of the normalized eigenvectors provide the weights. In the usual normalization,

$$\int f(x) d\mu(x) = \sum_{j=1}^n w_j f(\lambda_j) \quad (22.3)$$

for polynomials f of degree at most $2n-1$. This is the classical Golub–Welsch/Jacobi-matrix characterization.

22.1.2. Golub–Welsch extraction

For a dense set of all n nodes and weights from a symmetric tridiagonal Jacobi matrix, the classical Golub–Welsch route has $O(n^2)$ complexity. Faster specialised algorithms exist for particular families, but that is not the generic classical complexity assumed here.

22.2. The Toda lattice connection

For a Jacobi matrix the finite non-periodic Toda flow is the Lax equation

$$\dot{J} = [B, J], \quad B = J_+ - J_-, \quad (22.4)$$

with J_+ and J_- the strict upper and lower triangular parts. The flow changes Jacobi coefficients while preserving the spectrum. It therefore provides a deterministic isospectral reparameterisation, not a mechanism that “preserves the Jacobi coefficients” themselves.

A forward-Euler step with small η_{mix} is

$$J_{n+1} = J_n + \eta_{\text{mix}} [B(J_n), J_n] + O(\eta_{\text{mix}}^2). \quad (22.5)$$

22.3. The spectral observer principle

The programme's observer chain contains results of different evidence classes:

1. **Jacobi/Gaussian encoding (Class K).** The recurrence and quadrature relation in Equations (22.1) and (22.3) are classical.
2. **Isospectral Toda transport (Class K).** Exact Lax evolution preserves eigenvalues and trace invariants while evolving eigenvectors and recurrence coefficients.
3. **Divisor-state embedding (framework definition/results).** The angular metric is given by Chapter 14.
4. **CV-spectral relation (Class E).** Equation (11.2) is a fitted empirical law on the declared ensemble, not a theorem of random matrix theory.
5. **First-order Lax secular response (exact local calculation).** For $\Delta J = \eta[B, J]$, an eigenvector v_j of a simple eigenvalue satisfies $v_j^\top \Delta J v_j = 0$, so the first-order eigenvalue shift vanishes.
6. **Theta completion/oracle (Class O/E).** The theta-based completion remains conjectural, with the reported predictive benchmark treated as preliminary empirical validation.

22.3.1. Validation: the theta-oracle benchmark

The programme reports 30 randomized seeds and a mean lead time of 64.5 steps over a barycentric interpolation baseline. The baseline is structurally weak because it does not use the Lax pair. The result is therefore retained as preliminary validation only. Promotion requires comparison against at least one structure-aware extrapolator and one learned predictor, together with the full seed register and per-regime error bars (OP-9).

22.3.2. Candidate coordinate descriptions

The programme compares three coordinate descriptions:

Table 22.1.: *Candidate coordinate descriptions used in the spectral programme.*

Description	Coordinates	Associated system	Intended regime
Gauss–Christoffel	(λ_j, w_j)	Jacobi spectral theory	Baseline / all fibre entropies
Calogero–Moser	(q_j, p_j)	Rational CM model	Near repulsive transition
Ruijsenaars/Toda-type	(a_k, b_k)	Relativistic/Toda model	Strong-repulsion asymptotic

These coordinate descriptions are treated as candidate local models, not as a proved atlas on a single Grassmannian. A chart-compatibility theorem would require explicit transition maps, domains, and residual bounds.

23. Annular Spectral Waveguides and the Spectral Circuit IR

The spectral programme uses two classical operator geometries. Jacobi matrices encode orthogonal-polynomial recurrences for measures on the real line, while CMV matrices provide the canonical five-diagonal unitary representation for orthogonal polynomials on the unit circle. These are related in the classical literature, but not by a simple statement that an annulus collapses to one operator or the other. The standard OPUC–OPRL bridge is instead furnished by the Szegő mapping and the Geronimus relations, which connect suitable unit-circle measures and Verblunsky coefficients to real-line measures and Jacobi parameters.

The computational motivation remains legitimate: a spectral compiler that supports both Jacobi and CMV operations benefits from a common intermediate language. The annular spectral waveguide is therefore retained as a *conjectural unification target*, not as established orthogonal-polynomial theory.

23.1. Conjectural annular unification

Let

$$A(r_1, r_2) = \{z \in \mathbb{C} : r_1 \leq |z| \leq r_2\}, \quad 0 \leq r_1 \leq r_2. \quad (23.1)$$

Annular Spectral Waveguide Conjecture (Class O — OP-5). There may exist a family of spectral operators $W(r_1, r_2; \mu)$, together with explicitly declared limiting paths in the annular parameter space and corresponding measure maps, such that one limit reproduces a Jacobi operator $J(\mu_{\mathbb{R}})$ and another reproduces a CMV operator $C(\mu_{\mathbb{T}})$ in a stated mode of spectral convergence. A stronger version would also require the associated deformation laws and quadrature data to approach the Toda and defocusing Ablowitz–Ladik branches, respectively. These are conjectural requirements, not consequences of Equation (23.1).

Promotion of OP-5 requires four separate demonstrations:

- (a) construct $W(r_1, r_2; \mu)$ for at least one non-degenerate annulus and a concrete measure class;
- (b) state the two limiting paths and measure transformations explicitly, then verify convergence against known Jacobi and CMV examples;
- (c) define the annular spectral gap, contact-energy/readout quantities, and any proposed resource monotone, and establish their continuity or document the precise failure of continuity along the declared paths;
- (d) reproduce the same declared spectral observables across the reference implementation and supported numerical backends within fixed tolerances.

Until these conditions are met, the waveguide is an architectural conjecture. The known Szegő/Geronimus correspondence is a calibration target and prior-art constraint: any annular construction must be compatible with those classical relations where their hypotheses overlap, rather than rediscovering or contradicting them.

The conjecture is nevertheless useful to compiler design. A shared Spectral Circuit IR can expose separate JACOBI_STEP and CMV_STEP operations today while leaving room for a future, evidence-backed ANNULAR_STEP. That software affordance does not count as evidence for OP-5; it merely prevents the implementation architecture from assuming in advance that the two classical branches can never be unified.

23.2. The Spectral Circuit IR

The Spectral Circuit IR is a design-stage intermediate representation between spectral operations and executable backends. Its current schema names *thirteen* primitive operations.

Table 23.1. Primitive operations of the Spectral Circuit IR.

Operation	Status	Invariant / contract	Primary failure mode
INIT_STATE	Verified kernel	State dimension matches registry seed	Null payload or dimension mismatch
NORMALIZE	Verified kernel	$\ \psi\ = 1$ within tolerance	Tolerance violation
SPECTRAL_DECOMP	Verified kernel	Complete eigensystem; $\ Av - \lambda v\ < \epsilon$	Incomplete or inaccurate decomposition
JACOBI_STEP	Verified kernel	Tridiagonal structure; isospectral tolerance	Breakdown or spectral drift
CMV_STEP	Design / conjectural	Five-diagonal CMV form; $ \alpha_k < 1$	Invalid Verblunsky coefficient
CHEBYSHEV_FILTER	Verified kernel	Declared degree and passband tolerance	Gibbs ripple / degree inflation
TRANSPORT	Verified kernel	Norm preservation; declared inverse/adjoint	Unitary or reversibility drift
PROJECT	Verified kernel	$P^2 = P$ for projector branches	Rank/orthogonality mismatch
MEASURE	Design-stage IR op	Readout semantics and residual contract	False reversibility or missing loss accounting
ROUTE	Design-stage IR op	Target exists and compatibility contract passes	Incompatible pair / width below floor
COMPOSE	Verified rule	Type and evidence composition within tolerance	Associativity/type drift or evidence inflation
ASSERT_INVARIANT	Verified rule	Runtime invariant/failure-budget check	Stale definition or budget overrun
LOG_TRACE	Verified rule	Append-only schema-valid telemetry row	Write or schema failure

The lower-level kernels above may exist independently of the IR compiler; the statement that the IR is design-stage therefore does not imply that every numerical operation is unimplemented.

The most consequential semantic distinction concerns MEASURE. Suppose a normalized input $|\psi_0\rangle$ is transported forward, measured by a channel $\mathcal{E}$, and transported back to a normalized returned state $|\tilde{\psi}_0\rangle$. Define the round-trip fidelity and its logarithmic loss score by

$$F_{\text{rt}} = |\langle \psi_0 | \tilde{\psi}_0 \rangle|^2, \quad (23.2)$$

$$I_{\text{loss}} = -\ln F_{\text{rt}}. \quad (23.3)$$

This is a state-dependent fidelity-loss score. It is zero whenever the returned state equals the input up to phase; a non-trivial measurement channel may therefore yield zero on particular invariant states. It must not be interpreted as a channel-level proof of identity from a single input state.

A backend-equivalence gate may require a backend's evidence class to meet the operation's declared minimum class before lowering. That is a governance contract of the compiler architecture, not a mathematical consequence of the spectral theory.

23.3. The first canonical circuit

The first canonical circuit exercises the full operational cycle:

$$\begin{aligned} \text{INIT_STATE} \rightarrow \text{NORMALIZE} \rightarrow \text{SPECTRAL_DECOMP} \rightarrow \text{TRANSPORT_FORWARD} \\ \rightarrow \text{MEASURE} \rightarrow \text{TRANSPORT_REVERSE} \rightarrow \text{RESIDUAL} \rightarrow \text{LOG_TRACE}. \end{aligned} \quad (23.4)$$

INIT_STATE loads a registry state; NORMALIZE enforces the norm contract; SPECTRAL_DECOMP produces eigenvalues, eigenvectors, and quadrature weights; TRANSPORT_FORWARD applies the declared trajectory; MEASURE executes the readout; TRANSPORT_REVERSE applies the declared reverse/adjoint trajectory; RESIDUAL compares the returned and initial states; and LOG_TRACE writes the append-only record.

Interlude — Why Measurement First

There is a persistent temptation, in both mathematics and philosophy, to begin with ontology. One first decides what the furniture of the world must be, then designs languages and measurements that reveal it. The present work begins at the opposite end. It begins with the act of measurement and asks what survives that act under a declared protocol.

The resulting objects are thinner than the objects of a full ontology, but they are testable within their domain of competence. A measured state either coheres under perturbation or it does not. A residual is either within the declared tolerance or it is not. A gate either permits a transition or it does not. The deeper question of what exists independently of every measurement frame is left to other investigations.

This ordering also explains the treatment of quantum-like language. Spectral operators, chiral gradings, and uncertainty-shaped bounds appear because they arise in the mathematics of the constructed measurement geometry. Their formal resemblance to objects in quantum theory is useful; it is not a physical-ontology claim. The correction ledger in Appendix A is intended to make that restraint operational rather than rhetorical.

24. Operators, Rails, and Contact

Once measured states are available, one may ask how they move. The geometry admits isospectral dynamics that preserve eigenvalues while rotating the eigenvector frame. The Toda flow on Jacobi matrices is the elementary example. It is deterministic, classical, and exactly solvable in the appropriate coordinates. Casimir invariants are preserved. The flow can be viewed as a continuous change of measurement frame that leaves the underlying spectral measure intact. Contact geometry appears when one interprets constraint boundaries as rails in the spectral phase space. A rail is a curve that a measured state may touch under controlled dynamics. Contact is the regime in which the state is drawn toward the rail and thereafter constrained to move along it. Holonomy records the net transformation accumulated by parallel transport around closed loops. Clearance records the distance from forbidden regions. Failure modes are the geometric conditions under which contact is lost or rails are violated. The prime rail of Chapter 11 is the canonical example, and its name is not accidental. The rail is the contact structure of ordinal filtration: a measured state restricted to the prime subsequence is a state in sustained contact with the irreducibility boundary, and the relocation of complexity from value-space to index-space is what holonomy looks like on that rail. The enriched state construction is the parallel transport that keeps the valuesector and index-sector connected across the contact. The entire contact-mechanical layer is a reinterpretation of the earlier algebraic constraint cells as differential-geometric constraints on motion. The algebra is not discarded; it is given a dynamics. The same Spectral Observer-Eigenflow State remains the object that is moved, contacted, and gated. Computation, when it appears, will be understood as governed transformation of such states under residual accounting. The philosophical reflection is again one of restraint. The language of contact and rails is powerful and easily over-extended. The book uses it only where the geometry is under control and the residuals can be measured. Where the geometry is not yet under control, the language is withheld — as it is withheld, for instance, from the annular waveguide, whose contact structure cannot be drawn until the conjecture's demonstration (a) supplies a surface to draw it on.

25. Computation as Governed State Transformation

If the fundamental object is a measured state carrying residual and governance data, then computation is not the execution of an instruction stream against a passive memory. Computation is the controlled transformation of measured states under residual accounting. A step is admissible when the residual remains within tolerance and the governance gates permit release. A step is rejected when the residual exceeds tolerance or a gate closes. This view has immediate consequences for architecture.

Backends become interchangeable only when they preserve the residual structure. Attention mechanisms become routing operators that must themselves be residual-checked. Sequence models become trajectories of measured states rather than black-box function approximators. Fail-closed behaviour becomes the default: when residual accounting cannot be completed, the system does not guess; it refuses. The same view supplies a disciplined language for machine governance. Persistent identity across change is the survival of a Spectral Observer-Eigenflow State under admissible transformation. Promotion gates are the explicit conditions under which a candidate state is allowed to become active in a larger system. Ledgers are the appendonly history that makes later residual accounting possible. None of these notions requires a metaphysical theory of agency. They require only the measurement geometry already developed and the decision to treat residual accounting as non-optional. The philosophical stakes are visible. A computational system organised around residual-checked measured states is a system that can, in principle, know when it does not know. Whether any particular implementation achieves that standard is an empirical question. The standard itself is coherent and available — and the Spectral Circuit IR of Chapter 23 is its first concrete grammar, with MEASURE as the only irreversible word.

Part IV

General Theory, Branches, Evidence, and Synthesis

General theory supplies a typed grammar for local branches; it does not erase their differences.

26. General Theory of Perspective-Relative Structural Dynamics

Frame declaration. This chapter is the general theory layer. It is intentionally broader than any one arithmetic branch. Branch-specific laws enter only through typed slots; no branch is promoted merely because it fits the general schema.

The original SOES already couples raw object, frame, coordinates, derivative jet, defect vector, gates, and ledger. The present formulation makes the operator relations between those components explicit.

Definition 26.1 (Perspective-relative structural system). *A branch is represented by the typed tuple*

$$\mathfrak{B} = (X, \mathcal{F}, \Pi, \mu, \Phi, \mathcal{I}, \mathcal{N}, \mathcal{R}, \mathcal{G}, \mathcal{L}),$$

where:

X is the latent or pre-measurement state space;

$\mathcal{F}$ is the declared measurement frame;

$\Pi_{\mathcal{F}}$ is the observation/projection map;

μ is the typed interaction law;

Φ is the state-transition or transport dynamics;

$\mathcal{I}$ is the declared invariant family;

$\mathcal{N}$ is the null model;

$\mathcal{R}$ is residual and failure accounting;

$\mathcal{G}$ is governance/promotion logic;

$\mathcal{L}$ is the append-only evidence ledger.

The observed state is

$$y_t = \Pi_{\mathcal{F}}(x_t),$$

and a self-conditioned structural dynamic may take the form

$$x_{t+1} = \Phi_{\mathcal{F}}(x_t, S_{\mu}(x_t), c_t).$$

A branch need not use self-interaction; if it does not, the slot is null rather than silently overloaded.

26.1. Perspective transport

Given frames $\mathcal{F}$ and $\mathcal{G}$, a transport $T_{\mathcal{F} \rightarrow \mathcal{G}}$ is admissible only if its typing is declared. A strong compatibility relation is the commuting condition

$$\Pi_{\mathcal{G}} \circ T_{\mathcal{F} \rightarrow \mathcal{G}} = \hat{T}_{\mathcal{F} \rightarrow \mathcal{G}} \circ \Pi_{\mathcal{F}},$$

for some observable-space transport $\hat{T}$. When this fails, the mismatch is a residual, not a philosophical disagreement.

Similarly, an interaction-preserving map $\varphi : (X, \mu) \rightarrow (Z, \nu)$ satisfies

$$\varphi(\mu(x, y)) = \nu(\varphi(x), \varphi(y)),$$

which implies

$$\varphi \circ S_\mu = S_\nu \circ \varphi.$$

This is the precise substrate-agnostic criterion for preservation of the self-interaction branch.

26.2. Conserved quantity versus conserved representation

The general theory distinguishes four strengths of preservation:

Value preservation: one scalar observable is conserved.

Invariant-family preservation: a declared set $\mathcal{J}$ is conserved.

Operator preservation: the interaction law commutes with transport.

State-equivalence preservation: the transported state remains in the same certified equivalence class across the declared coordinate pentad and residual gates.

A lower level does not imply a higher one.

26.3. General theory and branch theory

The purpose of a general theory here is not to make every branch universal. It provides a common ledger of questions:

What is the state space?

What makes two states observationally equivalent?

What is the interaction law?

What dynamics transport the state?

Which invariants are supposed to survive?

Which projection creates the readout?

What lies in the measurement kernel?

What residual certifies failure?

What evidence would promote the claim?

Different branches may answer these questions differently and may be mutually inapplicable. The general layer requires typed boundaries, not forced unification.

26.4. Branch interaction

Two branches $\mathfrak{B}_1$ and $\mathfrak{B}_2$ may interact through a declared transducer

$$\mathcal{T}_{12} : Y_1 \rightarrow X_2$$

or through a common invariant object I . Composition is admissible only when the codomain of the first branch matches the domain expected by the second. A conceptual analogy is not a transducer.

26.5. Extension of the SOES derivative jet

Where the observable map depends differentiably on a parameterised projector or frame variable P , perspectival sensitivity can be represented by a Fréchet/directional jet. For an admissible perturbation δP , define

$$J_k^{(P)} A(a; \delta P) = \{D_P^j A(a; P)[(\delta P)^{\otimes j}] : 1 \leq j \leq k\}.$$

This notation requires a declared parameter space for P and differentiability of A with respect to that parameter; it is not defined for an arbitrary set-valued notion of “perspective”. This makes perspectival sensitivity first-class. A state may be dynamically stable yet perspectivally unstable; the defect vector should record those as distinct failure coordinates.

Bounds and portability. The tuple $\mathfrak{B}$ is a schema, not a theorem that every useful theory must adopt it. Its role is governance: every admitted branch in this volume can be reduced to these slots or must explicitly declare why it cannot.

27. Branches, Bounds, and Failure Modes

Frame declaration. Failure is typed. A claim may fail algebraically, observationally, dynamically, numerically, semantically, or by unauthorized promotion. These are different defects and require different remedies.

The present framework introduces a branch contract. Every nontrivial branch should declare the following before its results are interpreted.

Field	Required declaration
Domain	Raw objects/states admitted.
State type	Vector, matrix, fibre, graph, operator, spectrum, equivalence class, etc.
Interaction	Typed operator and arity.
Measurement plane	Observable/projector and normalisation.
Dynamics	Evolution or transport law, if any.
Invariants	Quantities claimed to survive admissible transformations.
Null model	Expected readout in absence of target structure.
Bounds	Parameter range and boundary conditions.
Failure modes	Conditions that invalidate inference or branch applicability.
Evidence tier	Proven, verified, supported, conjectured, falsified, or design-stage.
Portability bar	What must be re-derived before export to another substrate/frame.

27.1. Failure classes

Class	Failure	Diagnostic / containment
F-A Algebraic	Type mismatch, broken closure, false identity.	Exact counterexample; theorem audit; fail closed.
F-O Observability	Latent component lies in but is treated as absent.	Cross-probe or alternate plane; report equivalence class.
F-C Common mode	Universal/background component is mistaken for discriminative signal.	Center or quotient the common mode; compare planes.
F-I Interference	Set overlap is interpreted as signed interference without amplitudes.	Declare amplitude/phase algebra before interference language.
F-D Dynamics	A static structural identity is promoted to a law of motion.	Require an explicit and out-of-sample trajectory prediction.
F-S Symmetry	Degeneracy attributed to arithmetic structure when generated by accidental symmetry.	Random/local negative controls; perturbation splitting test.

Class	Failure	Diagnostic / containment
F-B Boundary	Cyclic law exported to open/non-periodic boundaries.	Boundary control; report leakage or failure of exact closure.
F-T Transport	Two charts/operators are assumed compatible although they do not commute.	Commutator, residual, sector leakage.
F-X Inverse	Microscopic parameter reported uniquely when only a quotient class is identifiable.	Resolution analysis ; equivalence-aware inference.
F-N Numeric	Roundoff/truncation is interpreted structurally.	Independent estimator, convergence sweep, append-only diagnostic.
F-E Evidence	PASS is treated as promotion authority.	Separate theorem, test, publication, and production gates.
F-P Physicalization	Internal measurement geometry is exported as physical ontology.	Re-derive under physical measurement protocol; preserve scope disclaimer.

27.2. Failure can itself be branch-local

A branch can fail while the general theory survives. Open boundaries falsify exact cyclic channel closure but do not falsify perspective-relative measurement. Random circulant kernels falsify generic divisor-channel degeneracy but do not falsify the exact-period classification. The distinction is essential: a general theory that can only survive by declaring every branch universal is brittle; a typed general theory expects local failure and records which dependency broke.

27.3. Negative controls as first-class evidence

The experimental programme uses three recurring negative controls:
ordinary nearest-neighbour coupling to test whether arithmetic channel degeneracy is generic;
random symmetric circulant kernels to test whether cyclicity alone causes the observed grouping;
open boundaries to test whether exact closure depends on periodic geometry.
A positive result without its corresponding negative control is incomplete evidence.

27.4. Promotion boundaries

The five-tier evidence discipline remains. This edition adds a second axis: **branch scope**. A result can be proven inside branch *B* while remaining inapplicable to branch *C*. Evidence tier answers “how well established?” Scope answers “where does it apply?” Both are required.

28. Experimental Programme and Falsifiable Predictions

Frame declaration. All experiments in this chapter are synthetic computational tests of declared model classes. A PASS validates the preregistered criterion for the implemented branch. It does not establish that an unrelated physical or learned system instantiates that branch.

The enrichment programme proceeded by freezing predictions before execution, preserving failures, and using negative controls designed to break the proposed structure when its assumptions were removed.

28.1. Evidence ladder

Stage	Scale	Principal result
Toy bilinear	operator	Diagonal probes reconstructed the symmetric component to about ; antisymmetric structure remained invisible; ordered cross-probes recovered the full operator.
Tiled operator	of blocks	Same diagonal observable with rank changing from 6 to 24; exact tile-space sector decomposition.
Arrangement/tile sweep	36 hierarchical cases	Channel count and shared periods matched exactly.
Large falsifiable suite	, 649,166 transitions	; symmetry-breaking and cross-scale leakage exponents 1 and 2.
Real-space batch	grids to	2D pure-mode law, open-boundary negative control, blind carrier prediction.
Twenty-family batch	20 preregistered families	20/20 met criteria, including 2D/3D fields, negative controls, inverse inference, and tiled compatibility.
High-range shadow sweep		smallest full resolver for ; first reaches under .

28.2. Major Batch 3 summary

ID	Family	Result
F01	Exhaustive channel census	PASS through ; maximum first at .
F02	1D nonlinear transitions	PASS, 350,000 random transitions.
F03	2D real-space transitions	PASS, 84 fields; up to 737,280 sites.
F04	3D real-space transitions	PASS, 30 fields; up to 368,640 sites.

ID	Family	Result
F05	Mixed-mode intermodulation	PASS; maximum forbidden energy fraction .
F06	Repeated general-order attractor	PASS, 120,000 cases.
F07	Semigroup composition	PASS, 400,000 cases.
F08	Arithmetic-kernel degeneracy	PASS; max within-channel dispersion .
F09	Random circulant control	PASS; exact arithmetic degeneracy 0/3976.
F10	Nearest-neighbour control	PASS; exact degeneracy 1/1020.
F11	Spectral symmetry breaking	PASS; split exponent 1.0000.
F12	Real-space disorder	PASS; amplitude exponent 1.0000, energy exponent 2.0000.
F13	Open-boundary control	PASS; periodic 6/6, open exact 0/6.
F14	Modular collision/aliasing	PASS, 20/20 deliberately colliding cases.
F15	Cross-scale overlap	PASS, 16,129 pairs.
F16	Broken cross-scale selection	PASS; amplitude 1.0000, energy 2.0000.
F17	Hidden operator-family identification	PASS, 100% on the synthetic held-out family set.
F18	Inverse order inference	PASS equivalence-aware 100%; unique only 8.6%.
F19	Blind mixed-spectrum prediction	PASS, mean precision=recall=1.000000.
F20	Tiled sector compatibility	PASS; sector error , mixing exponent 1.0000.

28.3. Falsifiable predictions retained

The following are suitable for blind tests because failure can be stated in advance.

Pure cyclic character under order- r pointwise self-interaction must move from exact period p to $p/\gcd(p, r)$.

Repeated order- r interaction must reach the factor of p coprime to r and must not remove unrelated prime factors.

A mixed-mode polynomial output must lie inside the modular additive closure of the input support.

Exact divisor-channel damping degeneracy must disappear under generic coupling that does not respect gcd-orbits.

A weak generic symmetry-breaking perturbation must split an exactly degenerate finite-dimensional branch at first order unless symmetry or cancellation removes the first-order term.

In the declared cross-scale channel model, forbidden amplitude must be first order and forbidden energy second order in weak coupling.

Open boundaries must destroy exact cyclic closure unless an independent structure recreates it.

Inverse inference must output the observational equivalence class when the measurement modulus cannot uniquely resolve the microscopic order.

28.4. The preregistered numerical failure retained

Major Batch 2 family E7 is preserved as a preregistered FAIL because its scalar leakage computation exceeded an unrealistically strict 10^{-18} threshold at approximately machine epsilon. A separate append-only diagnostic recomputed forbidden-bin energy directly and found a maximum of approximately

5.5×10^{-28} . The criterion was not retroactively edited. This is the intended use of the ledger: failure records the test contract; diagnosis records what the failure meant.

Bounds and portability. The 20/20 result is not a claim of 20 independent discoveries. Several families exercise exact identities or deliberately constructed symmetries. The scientific weight lies in the separation of theorem, implementation check, negative control, and blind prediction.

29. Reflection on Restraint

The book has now travelled from the elementary distinction between arithmetic and measurement to a geometry of residual-checked states and a view of computation as governed transformation. Along the way a series of refusals has been maintained. The zero mode has not been allowed to become a vacuum or a ground of being. Quantum-like structural features have not been allowed to become quantum ontology. Cultural and historical motifs have not been allowed to become priority claims. Failures have not been erased. Each refusal has a technical motive: the preservation of residual accountability. When a formal singularity is inflated into metaphysics, residual accounting becomes optional and the geometry loses its claim to precision. When structural resemblance is inflated into identity, the classical character of the constructions is obscured and false experimental expectations are generated. When history is inflated into priority, pattern sources are mistaken for evidence. This edition has had to practice the doctrine on itself. Three statements in an earlier formulation failed residual checks: a combinatorial justification (the partition miscount), a numerical value (the ideal tower at $s = 2$), and a flagship unification in its strong form (skip-set antipodality). Each correction is recorded in Appendix A with the counterexample or recomputation that forced it. The corrections do not weaken the book's thesis; they instantiate it. A measurement discipline that cannot correct its own measurements is a liturgy, and the whole point of the ledger is to prevent that slide. Restraint is therefore not an ethical decoration. It is the condition under which the positive claims remain answerable to residual checks. A geometry that cannot say what it is not cannot be trusted when it says what it is. The reader who has followed the argument is now in possession of a self-contained account. No other document is required. Further technical development is possible and welcome. Further philosophical inflation is not. The architecture is open at the end that faces new coordinates, sharper bounds and stronger computational realisations. It is closed at the end that faces totalising narrative. That combination is the practical meaning of the coherence the book has tried to practice as well as describe.

30. Synthesis: What Survives a Change of Perspective

The preceding chapters organized the programme around a measured state that survives declared perturbations of frame. The enrichment sharpens the question. A state can survive one frame while its shadow changes radically under another; a common mode can be signal in one task and nuisance in another; an invisible operator component can remain dynamically decisive despite lying in the kernel of a diagonal observer.

The general lesson is therefore not that every domain secretly obeys the same arithmetic. It is that a disciplined theory should separate four things that are often conflated:

latent state interaction law structural dynamics measurement plane.

The object of interest is not any one projection. It is the structured relation among these components and the invariants that survive admissible transport between them.

30.1. A compact general form

For a branch $\mathfrak{B}$, the operational cycle is

$$x_t \xrightarrow{\Delta/\mu} S_\mu(x_t) \xrightarrow{\Phi_{\mathcal{F}}} x_{t+1} \xrightarrow{\Pi_{\mathcal{F}}} y_{t+1} \xrightarrow{\mathcal{R}} (\text{residual, defect, evidence}).$$

Not every branch uses every arrow. The graph is a type system, not a demand for superficial uniformity.

30.2. The arithmetic branch as an example, not a universal template

In the cyclic divisor-channel branch:

exact periods are divisors;

interaction order acts by $p \mapsto p / \gcd(p, r)$;

shared cross-scale channels are divisors of the gcd;

divisor-state overlap is $\tau(\gcd(a, b))$;

operator resolution is governed by gcd fingerprints;

prime shadows carry a universal common-mode half-overlap on the source-normalized divisor plane.

These are exact or experimentally supported statements under the branch's definitions. They are not licenses to label every periodic pattern, every prime-like object, or every spectral degeneracy with the same vocabulary.

30.3. What is conserved when perspective changes?

The strongest candidate for substrate-agnostic structure is whatever commutes with admissible transport:

$$\hat{T}_{\mathcal{F} \rightarrow \mathcal{G}} \circ \mathcal{O}_{\mathcal{F}} \approx \mathcal{O}_{\mathcal{G}} \circ T_{\mathcal{F} \rightarrow \mathcal{G}},$$

within declared residual tolerance. The conserved object may be a scalar invariant, an equivalence class, an operator relation, a topology, or a causal dependency. The representation itself is expendable unless it is one of the declared invariants.

This suggests a practical research programme. For every proposed concept, build at least two genuinely different measurement planes, transport the object between them, and ask which claims survive. If a claim disappears, record the kernel that swallowed it. If it changes sign, phase, or type, record the transducer. If it survives, attempt to prove the commutation relation that made survival possible.

30.4. Restraint retained

The integrated framework expands the architecture but does not relax its restraint. A branch may be powerful and narrow. A synthetic experiment may be predictive and still not physical. A general theory may organize many branches without asserting that the branches are ontologically identical. The burden of transfer remains re-derivation under the target frame.

The updated thesis can therefore be stated compactly:

An object is what survives measurement under a declared frame. A shadow is what that object becomes under a declared projection. A structural law earns portability only when its invariants survive transport between frames within declared residual bounds.

The programme remains open at the end that faces new coordinates, sharper transport laws, and better falsification. It remains closed against the move from structural resemblance to universal identity.

Epilogue — Closing Reflection

The book has moved from a distinction between operation and measurement to a geometry of residual-checked states, and from that geometry to operators, towers, contact structures, and governed computation. The same discipline has been visible throughout: state, frame, observable, residual, evidence, and scope must remain distinguishable.

Two commitments are retained. The first is operational: a candidate object is recognized through what remains coherent under a declared measurement frame and its admissible perturbations. The second is restraint: structural resemblance to quantum formalism is not quantum ontology; a zero mode is not a physical vacuum; a synthetic benchmark is not a theorem; and a PASS is not a downstream authority grant.

This volume is sufficient to follow the definitions, evidence classes, and logical dependencies of the argument. Reproduction of empirical claims still requires the corresponding computational harnesses, seeds, tolerances, and raw evidence. That distinction is deliberate. The architecture remains open to richer coordinates, sharper residual bounds, and stronger computational realisations, while remaining closed against the overclaims that would erase its competence boundaries.

Ad astra, nulle nomen.

Appendices

A. Revision, Errata, Falsification, and Enrichment Ledger

Appendix A — Revision and Errata Ledger. The ledger is append-only. Each entry records the earlier statement, the defect, the correction, and the evidence that forced it. Falsifications are infrastructure, not confession.

Table A-1. Errata corrected in the revised edition.

ID	Earlier statement	Defect	Correction	Forcing evidence
E-1	“ $n+1$ counts partitions of n into at most two parts.”	Combinatorial miscount	Family renamed composition fibre H_{comp} ; $n+1$ counts ordered pairs (a, b) with $a+b=n$	Direct enumeration: $n=4$ gives three partitions but five ordered non-negative compositions
E-2	IG-T03: $\gcd(n, m) = 1 \Leftrightarrow d_F(n, m) = \pi$; related QI-T01 claim	Uniform divisor states always share divisor 1	Refined IG-T03’: coprimality $\Leftrightarrow$ minimal overlap $\{1\} \Leftrightarrow$ zero multiplicative entropy production $\Leftrightarrow$ exact tensor factorisation	Counterexample $(4, 9)$ gives $d_F \approx 2.85 < \pi$; analytic overlap $1/\sqrt{d(n)d(m)} > 0$
E-3	Ideal tower for $Q(i)$ at $s=2$ printed as ≈ 2.008	Numerical error	$\zeta(2)L(\chi_{-4}, 2) \approx 1.5067$	Analytic evaluation and direct summation
E-4	Shared prime factors produce positive multiplicative entropy	Sign contradiction	Non-coprime pairs produce a strictly negative deficit; zero occurs exactly for coprime pairs	$d(nm) \leq d(n)d(m)$; e.g. $\Delta S_{\text{mul}}(2, 6) = \ln(3/4)$

Table A-2. Falsifications retained in the ledger.

ID	Claim	Status	Note
F-04	GOE transition at $H_* = 12$	Falsified	Transition located in $(\ln 6, \ln 7)$
F-06	β_{eff} driven by $H_{\text{fact}}(N)$	Falsified	CV is the stronger driver ($r = +0.905$ versus $+0.534$); subsumed by T-CV1
F-07	Measurement residuals evidence spontaneous time-crystal structure	Falsified	No subharmonic response in 1,600 driven configurations; period-1 $\rightarrow$ aperiodic $\rightarrow$ divergence with no period doubling

Table A-3. Artefact post-mortems retained in the ledger.

ID	Artefact	Root cause	Lesson
Art-1	Round-trip residuals “scaling with $d(n)$ ” as a complexity law	$O(1/k)$ truncation error from a hand-built integrator; fit only $R^2 = 0.86$	Genuine geometric phases fit power laws tightly; artefacts smear
Art-2	Construction-dependent reversibility boundaries	Flow saturation on unnormalised matrices with mismatched spectral norms	Convergence control precedes interpretation
Art-3	Attempted dynamical breaking of reversibility	Blocked by CD-T06: autonomous flows are reversible by group property	The arrow of time lives in the measurement axis

Calibration changes (not errors): the theta-oracle benchmark is re-presented as preliminary validation with an explicit promotion bar (OP-9); the term “Landauer floor” is retained with an explicit non-physicality disclaimer (Ch. 4); the two phase-continuum narratives (entropy domain vs. phase-disorder domain) are explicitly separated (Ch. 15); the duplicated three-proofs material of an earlier formulation §§10–12 is consolidated into Chapters 14 and 16; and a standing scope qualification — every claim is internal to this one system, not mathematics in general — is now stated in the Note on Reading, the Contribution Statement, and at the highest-risk passages (Ch. 10, 11).

ID	Addition / refinement	Status
V17-01	Self-interaction defined as diagonal restriction .	Framework definition.
V17-02	Diagonal observability separates symmetric reconstruction from antisymmetric kernel.	Classical theorem + internal interpretation.
V17-03	Perspective operator and shadow formalized as .	Framework definition.
V17-04	Exact-period channel classification ; multiplicity .	Class K, verified computationally.
V17-05	Nonlinear channel action and semigroup law.	Proven from cyclic characters; verified computationally.
V17-06	Mixed-mode modular additive closure.	Algebraic support law; numerically stress-tested.
V17-07	Cross-scale exact-period overlap .	Class K + branch interpretation.
V17-08	Arithmetic shadow and normalized shadow .	Exact in divisor frame.
V17-09	Operator equivalence .	Proven within nonlinear channel branch.
V17-10	Resolving power and high-range capacity sweep.	Definition + numerical survey.
V17-11	Branch contract and failure-class taxonomy.	Governance framework.
V17-12	E7 preregistered numerical-threshold failure retained with append-only diagnostic.	Falsification/calibration record.

B. Glossary of Canonical Names

Appendix B — Glossary of Canonical Names. One name is used per object throughout this edition.

Table B-1. Canonical names and definitions.

Canonical name	Symbol	Definition	First defined
Arithmetics	—	Measurement of arithmetic objects under declared frames	Ch. 1
Measurement frame	F	Domain, units, observables, coordinate chart, null model, and governance rules	Ch. 1
Constraint cell	$C = (X, E, \pi, D)$	Solution space, constraints, projection, and domain declaration	Ch. 2
Zero mode	—	Identity-like degeneracy of the coadjoint orbit at radius zero	Ch. 4
Minimal fibre entropy	$\lambda_{\min} = \ln 2$	Fibre entropy of any prime; “Landauer floor” by analogy of form only	Ch. 4
Spectral Observer–Eigenflow State	$\text{SOES}(a)$	Canonical seven-component measured state	Ch. 7
Eigenform Coordinate Pentad	$A(a)$	Angular, functorial, informational, monoidal, and temporal coordinates	Ch. 7
Cellular Integer Framework	CIF	Nine-layer substrate architecture	Ch. 8
Factorisation fibre entropy	$H_{\text{fact}}(n) = \ln d(n)$	Logarithm of the positive-divisor count	Ch. 9
Composition fibre entropy	$H_{\text{comp}}(n) = \ln(n + 1)$	Logarithm of the count of ordered two-compositions	Ch. 9
Coloured Integer Matrix	CIM	Integer matrix on a complete k -partite graph with zero diagonal blocks	Ch. 8
Arithmetic Eigenform Engine state	$\text{AEE}(n) = \bigotimes_p \text{Sym}^{a_p}(C_2)$	Representation-theoretic state associated with n	Ch. 21
Prime rail	$P(n) = p_n$	Primes under ordinal re-indexing	Ch. 11
Superprime rail	$P^2(n) = p_{p_n}$	Primes whose prime-sequence index is itself prime	Ch. 11
Enriched prime-rail state	$\text{AEE}^+(p_n) = C_2 \otimes \text{AEE}(n)$	Value-sector and index-sector state	Ch. 11
Divisor coefficient of variation	$\text{CV}(N) = \sigma_N / \mu_N$	Spread of the divisor set; control parameter for level repulsion in the declared ensemble	Ch. 11
Quantum number state	$ \psi_N\rangle$	Uniform state over divisors of N in ℓ^2	Ch. 14
Fisher distance	d_F	Twice the arccosine of divisor-state overlap	Ch. 14
Skip set	$S = \{(n, m) : \gcd(n, m) = 1\}$	Zero-loss locus for multiplicative entropy production	Ch. 14

Canonical name	Symbol	Definition	First defined
Readout operator	P_ε	$\text{diag}(J) + (1 - \varepsilon)\text{off}(J)$; Cartan projection at $\varepsilon = 1$	Ch. 14
Round-trip residual	R_{rt}	Residual after forward flow, readout, and reverse flow	Ch. 14
Holonomy budget	B_H	Off-diagonal Frobenius energy	Ch. 15
Operator-constraint morphospace	$\mathcal{K}$	Typed eight-axis space of computational species	Ch. 17
Computational prime	—	Irreducible stable attractor in $\mathcal{K}$ under a declared factorisation operation	Ch. 17
Automorphic ladder	$L_0 \rightarrow \cdots \rightarrow L_\infty$	Fibre Dirichlet-series ladder under $\partial_r _{r=0}$	Ch. 21
Jacobi chassis	$J(n)$	Divisor-indexed symmetric tridiagonal operator	Ch. 22
Annular spectral waveguide	$W(r_1, r_2; \mu)$	Conjectural operator family interpolating Jacobi and CMV limits	Ch. 23
Spectral Circuit IR	—	Thirteen-operation intermediate representation for spectral circuits	Ch. 23

C. Canonical Notation and Branch-Contract Schema

Table C.1. Canonical notation for the perspective-relative branches.

Symbol	Meaning
$\Delta_X : x \mapsto (x, x)$	Diagonal map used to construct self-interaction.
$S_\mu = \mu \circ \Delta_X$	Self-interaction induced by a typed interaction law μ .
$\Pi_{\mathcal{F}}$	Measurement/projection operator associated with frame $\mathcal{F}$.
$y = \Pi_{\mathcal{F}}(x)$	Projected state in the declared measurement frame.
$p_n(k) = n / \gcd(n, k)$	Exact closure period of a cyclic character mode.
$T_r(p) = p / \gcd(p, r)$	Order- r divisor-channel transition map.
$O(a, b) = d(\gcd(a, b))$	Unnormalised divisor-shadow overlap count.
$\Omega(a \rightarrow b) = d(\gcd(a, b)) / d(a)$	Source-normalised divisor shadow.
$\mathfrak{B}$	Typed perspective-relative structural branch.
$T_{\mathcal{F} \rightarrow \mathcal{G}}$	Declared transport between measurement frames.
$\mathcal{R}$	Residual and failure accounting.

C.1. Canonical branch contract

Every branch should be registrable as a machine-readable record with at least:

```

branch_id
version
state_space
frame
observable_or_projector
interaction_operator
transport_or_dynamics
invariants
null_model
parameter_bounds
boundary_conditions
failure_modes
evidence_tier
portability_bar
supersedes / superseded_by
experiment_or_proof_refs

```

This schema is intentionally compatible with the append-only ledger discipline of the SOES.

D. Register of Open Problems and Promotion Bars

Appendix D — Register of Open Problems. The table states the current open status and the promotion bar where one is already defined.

Table D-1. Open problems and promotion bars.

ID	Problem	Status / promotion bar
OP-1	Automorphic ladder top rung L_∞ : a Langlands-type L -function unifying the lower rungs	Open; a validated explicit CIM spectral-zeta zero locus remains to be derived
OP-2	Express the Gaussian Euler correction $Q_D(s)$ in standard L -function terms	Open
OP-3	Novelty in higher prime-rail iterates $P^k, k \geq 3$	Open; heatmap evidence presently suggests regularisation only
OP-4	Critical exponent $\nu \approx 0.262$: coincidence with $1/4$ or shared universality?	Open; no physical-criticality claim
OP-5	Annular spectral waveguide conjecture	Open; promotion bar (a)–(d) in Ch. 23; literature check against Simon’s OPUC required
OP-6	Theta-function gap-collapse oracle as an exact algebraic formula	Conjecture; numerical correspondence only for $n \leq 12, k \leq 3$
OP-7	Derive the phase-continuum exponent $\nu_{\text{phase}} = 0.21$ rather than fit it	Open
OP-8	Separate chromodynamic eigenflow curvature from the easy spectral-arithmetic subtheory	Open; promotion criterion 3
OP-9	Promote the theta oracle against structure-aware baselines, with published harness and per-regime error bars	Open; current result is preliminary validation
OP-10	Computational-prime certification of Eigenalgebra Arithmetics	Candidate; criteria 3, 6, and 7 remain open
OP-11	Morphospace compositionality: do two Eigenalgebra systems compose while preserving all eight axes?	Open
OP-12	Systematic enumeration of VC–VI rail states	Conjectured population; no complete enumeration

D.1. Framework Extensions

The following entries extend the earlier open-problem register and inherit the same evidence discipline.

ID	Problem	Promotion bar
OP-13	General classification of interaction laws recoverable from diagonal probes beyond symmetric multilinear forms.	Characterize diagonal kernel and minimal cross-probe set.
OP-14	Nonlinear perspective transport between incompatible measurement planes.	Construct typed with residual bounds and counterexamples.
OP-15	Complete arithmetic-channel perturbation theory beyond first order.	Derive second-order splitting and identify symmetry-protected cancellations.
OP-16	Blind operator-family identification against structure-aware spectral baselines.	Pre-register train/test split and beat Fourier/state-space baselines out of family.
OP-17	Optimal resolving modulus for candidate operator families above .	Prove or bound minimum achieving target or uniqueness fraction.
OP-18	Phase-sensitive divisor shadows and genuine destructive interference.	Declare complex channel amplitudes and demonstrate reproducible cancellation not reducible to counting.
OP-19	Non-cyclic and open-boundary analogues of divisor channels.	Identify replacement taxonomy or prove no exact analogue exists.
OP-20	Coupled branch calculus for arithmetic channels plus Jacobi/Toda dynamics.	Derive transducer and test whether channel labels predict spectral-flow observables beyond baseline.
OP-21	Higher-dimensional tiled sector compatibility under non-normal operators.	Generalize commuting-sector test and quantify pseudospectral failure.
OP-22	Perspective-aware extension of the SOES defect vector.	Implement projector derivatives and certify residual semantics.
OP-23	Branch registry and executable frame contracts in Spectral Circuit IR.	Compile at least three incompatible branches with static type rejection.
OP-24	Empirical transfer to learned sequence models.	Pre-register black-box probes; test whether arithmetic channel predictions explain held-out dynamics better than generic spectral features.
OP-25	Formal proof package for the arithmetic-channel theorems.	Lean formalization of census, transition, semigroup, overlap, and equivalence theorems with explicit mappings.

D.2. Closing note

The enlarged programme is allowed to branch. Its discipline is not universal applicability but explicit locality: every branch says where it lives, how it moves, what it measures, what can hide from that measurement, how it can fail, and what would be required to transport it elsewhere.

E. References

The following classical and external sources are cited for standard results, terminology, or calibration values. Framework-specific theorems and empirical claims remain identified by their internal IDs and evidence classes.

- Ablowitz, M. J., & Ladik, J. F. (1976). A nonlinear difference scheme and inverse scattering. *Studies in Applied Mathematics*, 55(3), 213–229. <https://doi.org/10.1002/sapm1976553213>.
- Amari, S.-i. (2016). *Information Geometry and Its Applications*. Springer.
- Apostol, T. M. (1976). *Introduction to Analytic Number Theory*. Springer.
- Atas, Y. Y., Bogomolny, E., Giraud, O., & Roux, G. (2013). Distribution of the ratio of consecutive level spacings in random matrix ensembles. *Physical Review Letters*, 110, 084101. <https://doi.org/10.1103/PhysRevLett.110.084101>.
- Cantero, M. J., Moral, L., & Velázquez, L. (2003). Five-diagonal matrices and zeros of orthogonal polynomials on the unit circle. *Linear Algebra and its Applications*, 362, 29–56. [https://doi.org/10.1016/S0024-3795\(02\)00457-3](https://doi.org/10.1016/S0024-3795(02)00457-3).
- Flaschka, H. (1974). The Toda lattice. II. Existence of integrals. *Physical Review B*, 9(4), 1924–1925. <https://doi.org/10.1103/PhysRevB.9.1924>.
- Fulton, W., & Harris, J. (1991). *Representation Theory: A First Course*. Springer.
- Gautschi, W. (2004). *Orthogonal Polynomials: Computation and Approximation*. Oxford University Press.
- Golub, G. H., & Meurant, G. (2010). *Matrices, Moments and Quadrature with Applications*. Princeton University Press.
- Golub, G. H., & Welsch, J. H. (1969). Calculation of Gauss quadrature rules. *Mathematics of Computation*, 23(106), 221–230. <https://doi.org/10.1090/S0025-5718-69-99647-1>.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183–191. <https://doi.org/10.1147/rd.53.0183>. Cited here for a naming analogy only; no physical Landauer bound is asserted.
- Simon, B. (2005). *Orthogonal Polynomials on the Unit Circle* (Vols. 1–2). American Mathematical Society.
- Teschl, G. (2000). *Jacobi Operators and Completely Integrable Nonlinear Lattices*. American Mathematical Society.
- Toda, M. (1967). Vibration of a chain with nonlinear interaction. *Journal of the Physical Society of Japan*, 22(2), 431–436. <https://doi.org/10.1143/JPSJ.22.431>.
- Weyl, H. (1939). *The Classical Groups: Their Invariants and Representations*. Princeton University Press.

